MILITARY SPECIFICATION

MODULES, DIGITAL, STANDARD ELECTRONIC

STATIC RANDOM ACCESS MEMORY MODULE, KEY CODE DGG

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the modules described herein shall consist of this specification and the latest issue of MIL-M-28787.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the acquisition of an electronic static Random Access Memory (RAM) module. This module is class I as specified in MIL-STD-1389.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, Code SEA-55Z3, Department of the Navy, Washington, D.C. 20362-5101, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specification, standards and handbook. The following specification, standards and handbook form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.

MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program.

HANDBOOK

MILITARY

MIL-HDBK-217 - Reliability Prediction of Electronic Equipment.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 Qualification. Modules furnished under this specification shall be products which are authorized by the qualifying activity for listing on the applicable qualified products list by the time of award of contract (see 4.5).

3.2 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

3.3 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

3.4 Failure rate requirements. All modules procured to this specification shall have a failure rate predicted in accordance with MIL-STD-1389. The predicted failure rate shall not exceed 6.6700 failures per million operating hours when predicted in accordance with DOD-HDBK-217E.

3.5 Performance requirements. The module shall conform to the electrical requirements specified in table I and shall operate within the environment of design class I as specified in MIL-STD-1389.

3.6 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.6.1 Key code. The module key code shall be DGG.

3.6.2 Size. The module size shall be 2A Format B.

3.6.3 Weight. The module shall have a maximum weight of 130 grams.

3.6.4 Contact pins. The module shall have 100 contact pins positioned 1 through 50 in one row and 51 through 100 in the other row.

3.6.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

3.6.6 Depth. The module shall have no additional depth requirements.

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TABLE I. Module electrical requirements.

Test cond	Parameter 5/	Ref test table	Test type 1/-3/	25 ± 5°C 4/		0 ± 5°C 4/		80 ± 5°C 4/		Units
				Min	Max	Min	Max	Min	Max	
A1	Isolation, initial	II	S	10	INF	10	INF	10	INF	MOhm
	end-of-life	II	S	1	INF	1	INF	1	INF	MOhm
A2-A25	Isolation, initial	II	X	10	INF	10	INF	10	INF	MOhm
	end-of-life	II	X	1	INF	1	INF	1	INF	MOhm
B1-B5	Continuity	III	X	0.0	1.0	0.0	1.0	0.0	1.0	Ohm
B6-B10	Continuity	III	Q	0.0	0.1	0.0	0.1	0.0	0.1	Ohm
C	Vcc stress	IV	S	----------	Module shall pass all remaining tests					-------
D1-D11	Ibd	V	S	0.0	0.1	0.0	0.1	0.0	0 1	mA
D12	Ibd	V	S	0.0	1.0	0.0	1.0	0 0	1 0	mA
D13-D28	Ibd	V	S	0.0	0.1	0.0	0.1	0.0	0.1	mA
D29	Ibd	V	S	0.0	1.0	0.0	1.0	0 0	1 0	mA
D30	Ibd	V	S	0.0	1.1	0.0	1.1	0.0	1 1	mA
D31	Ibd	V	S	0.0	0.2	0.0	0.2	0.0	0.2	mA
D32-D34	Ibd	V	S	0.0	1.1	0.0	1.1	0.0	1 1	mA
D35,D36	Ibd	V	S	-0.01	0.111	-0.01	0.111	-0.01	0.111	mA
D37-D44	Ibd	V	S	0.0	0.1	0.0	0.1	0.0	0.1	mA
D45	Ibd	V	S	-0.1	1.071	-0.1	1.071	-0.1	1.071	mA
E1-E29	Vic	VI	S	-1.5	-0.4	-1.5	-0.4	-1.5	-0.4	V
E30	Vic	VI	S	-1.2	-0.4	-1.2	-0.4	-1.2	-0.4	V
E31-E42	Vic	VI	S	-1.5	-0.4	-1.5	-0.4	-1.5	-0.4	V
F1-F16	Ios	VII	S	-225.0	-40 0	-225.0	-40.0	-225.0	-40.0	mA
F17-F22	Ios	VII	S	-110.0	-15.0	-110.0	-15.0	-110.0	-15.0	mA
F23,F24	Ios	VII	S	-100.0	-40.0	-100.0	-40.0	-100.0	-40.0	mA
F25	Ios	VII	S	-110.0	-15.0	-110.0	-15.0	-110.0	-15.0	mA
F26,F27	Ios	VII	S	-130.4	-30.03	-130.4	-30.03	-130.4	-30.03	mA
F28,F29	Ios	VII	S	-130.8	-30.06	-130.8	-30.06	-130.8	-30.06	mA
F30,F31	Ios	VII	S	-100.0	-15.0	-100.0	-15.0	-100.0	-15.0	mA
F32,F33	Ios	VII	S	-100.25	-15 0	-100.25	-15.0	-100.25	-15.0	mA
F34,F35	Ios	VII	S	-100.4	-15 03	-100.4	-15.03	-100.4	-15.03	mA

- See footnotes at end of table.

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TABLE I. Module electrical requirements - Continued.

Test cond	Parameter 5/	Ref test table	Test type 1/-3/	25 ± 5°C 4/		0 ± 5°C 4/		60 ± 5°C 4/		Units
				Min	Max	Min	Max	Min	Max	
F36	Ios	VII	S	-110.4	-15.0	-110.4	-15.0	-110.4	-15.0	mA
F37	Ioe	VII	S	-102.4	-15.0	-102.4	-15.0	-102.4	-15.0	mA
F38	Ios	VII	S	-100.0	-40.0	-100.0	-40.0	-100.0	-40.0	mA
G1-G11	Iih	VIII	X	0.0	20.0	0.0	20.0	0.0	20.0	uA
G12	Iih	VIII	X	0.0	25.0	0.0	25.0	0.0	25.0	uA
G13-G28	Iih	VIII	X	0.0	20.0	0.0	20.0	0.0	20.0	uA
G29	Iih	VIII	X	0.0	50.0	0.0	50.0	0.0	50.0	uA
G30	Iih	VIII	X	0.0	70.0	0.0	70.0	0.0	70.0	uA
G31	Iih	VIII	X	0.0	40.0	0.0	40.0	0.0	40.0	uA
G32-G34	Iih	VIII	X	0.0	45.0	0.0	45.0	0.0	45.0	uA
G35,G36	Iih	VIII	X	-310.0	-200.0	-310.0	-200.0	-310.0	-200.0	uA
G37-G44	Iih	VIII	X	0.0	20.0	0.0	20.0	0.0	20.0	uA
G45	Iih	VIII	X	-3.04	-2.35	-3.04	-2.35	-3.04	-2.35	mA
H1-H11	Iil	IX	X	-400.0	-30.0	-400.0	-30.0	-400.0	-30.0	uA
H12	Iil	IX	X	-250.0	0.0	-250.0	0.0	-250.0	0.0	uA
H13-H28	Iil	IX	X	-200.0	0.0	-200.0	0.0	-200.0	0.0	uA
H29	Iil	IX	X	-2.0	0.0	-2.0	0.0	-2.0	0.0	mA
H30	Iil	IX	X	-2.4	-0.03	-2.4	-0.03	-2.4	-0.03	mA
H31	Iil	IX	X	-800.0	-60.0	-800.0	-60.0	-800.0	-60.0	uA
H32-H34	Iil	IX	X	-650.0	-30.0	-650.0	-30.0	-650.0	-30.0	uA
H35,H36	Iil	IX	X	-921.1	-488.6	-921.1	-488.6	-921.1	-488.6	uA
H37,H38	Iil	IX	X	-370.0	-120.0	-370.0	-120.0	-370.0	-120.0	uA
H39-H44	Iil	IX	X	-400.0	-30.0	-400.0	-30.0	-400.0	-30.0	uA
H45	Iil	IX	X	-9.43	-5.0	-9.43	-5.0	-9.43	-5.0	mA
J1-J16	Voh	X	X	2.4	4.5	2.4	4.5	2.4	4.5	V
J17-J21	Voh	X	X	2.5	4.5	2.5	4.5	2.5	4.5	V
J22-J33	Voh	X	X	2.4	4.5	2.4	4.5	2.4	4.5	V
J34-J40	Voh	X	X	2.5	4.5	2.5	4.5	2.5	4.5	V
K1-K21	Vol	XI	X	0.0	0.4	0.0	0.4	0.0	0.4	V
K22,K23	Vol	XI	X	0.0	0.5	0.0	0.5	0.0	0.5	V
K24-K27	Vol	XI	X	0.0	0.4	0.0	0.4	0.0	0.4	V

- See footnotes at end of table.

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TABLE I. Module electrical requirements - Continued.

Test cond	Parameter 5/	Ref test table	Test type 1/-3/	25 ± 5°C 4/		0 ± 5°C 4/		80 ± 5°C 4/		Units
				Min	Max	Min	Max	Min	Max	
K28,K29	Vol	XI	X	0.0	0.5	0.0	0.5	0.0	0.5	V
K30,K31	Vol	XI	X	0.0	0.4	0.0	0.4	0.0	0.4	V
K32,K33	Vol	XI	X	0.0	0.5	0.0	0.5	0.0	0.5	V
K34-K39	Vol	XI	X	0.0	0.4	0.0	0.4	0.0	0.4	V
K40	Vol	XI	X	0.0	0.5	0.0	0.5	0.0	0.5	V
L1-L16	Iozh	XII	X	-20	20	-20	20	-20	20	uA
M1-M16	Iozl	XIII	X	-200	200	-200	200	-200	200	uA
N	Icc	XIV	X	81.8	818.5	81.8	818.5	81.8	818.5	mA
P1,P2	Functional	XV	X	----------		See table XV		----------		
Q1,Q2	Dynamic	XVI	S	----------		See table XVI		----------		
Q3,Q4	Dynamic	XVI	Q	----------		See table XVI		----------		
R	PSFC	XVIII	S	3.78	7.02	3.78	7.02	3.78	7.02	uF
S	PSTA	XIX	Q	0	400	0	400	0	400	mV
T1-T5	Output crosstalk (high level)	XX	Q	0	400	0	400	0	400	mV
U1-U5	Output crosstalk (low level)	XXI	Q	0	400	0	400	0	400	mV
V1-V12	Cin	XXII	Q	2	30	2	30	2	30	pF
V13-V38	Cin	XXII	Q	2	60	2	60	2	60	pF
V39-V44	Cin	XXII	Q	2	30	2	30	2	30	pF
V45	Cin	XXII	Q	2	75	2	75	2	75	pF

- See footnotes at end of table.

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TABLE I. Module electrical requirements - Continued.

Test cond	Parameter 5/	Ref test table	Test type 1/-3/	25 ± 5°C 4/		0 ± 5°C 4/		60 ± 5°C 4/		Units
				Min	Max	Min	Max	Min	Max	
W1-W20	Cout	XXIII	Q	0	60	0	60	0	60	pF
W21-W24	Cout	XXIII	Q	0	30	0	30	0	30	pF
X1-X32	Disable time	XXIV	Q	5	88	5	88	5	88	nS
X33-X64	Disable time	XXIV	Q	4	74	4	74	4	74	nS
Y	Threshold	XXV	X	----------		----------	See table XXV	----------		

- See footnotes on next page.

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1/ "X" parameters shall be 100 percent tested during the module acceptance test.

2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.

3/ "Q" parameters shall be tested during qualification testing only.

4/ Unless otherwise specified in the test table, limits are for initial and end-of-life.

5/ Abbreviations used in table I are defined in 6.1.

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3.6.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

3.6.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part or identifying number M28787/366-1. No marking shall interfere with heat dissipation from the fin (such as affixing a label which may act as a heat barrier).

3.7 Circuit pin configuration. The functional block diagram shall conform to figure 1. The pin assignments shall conform to figure 2.

3.8 Acceptance test. All production modules shall be tested in accordance with footnotes 1 and 2 in table I.

3.9 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits in table I (see 6.2).

3.10 Power supply voltage. The module shall be capable of conforming to the requirements in table I for a power supply voltage (Vcc) of 5.0 ± 0.5 V dc, except the output short circuit current limits apply only for a power supply voltage of 5.5 V dc ± 1 percent.

3.11 Parallel outputs. Paralleling of outputs is not permissible.

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

4.3 Test equipment. Test equipment specified on figures and in tables with a designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. Unless otherwise specified herein, all resistors shall be metal film and all capacitors shall be ceramic.

4.4 Test procedures. The module shall be tested as specified in table I.

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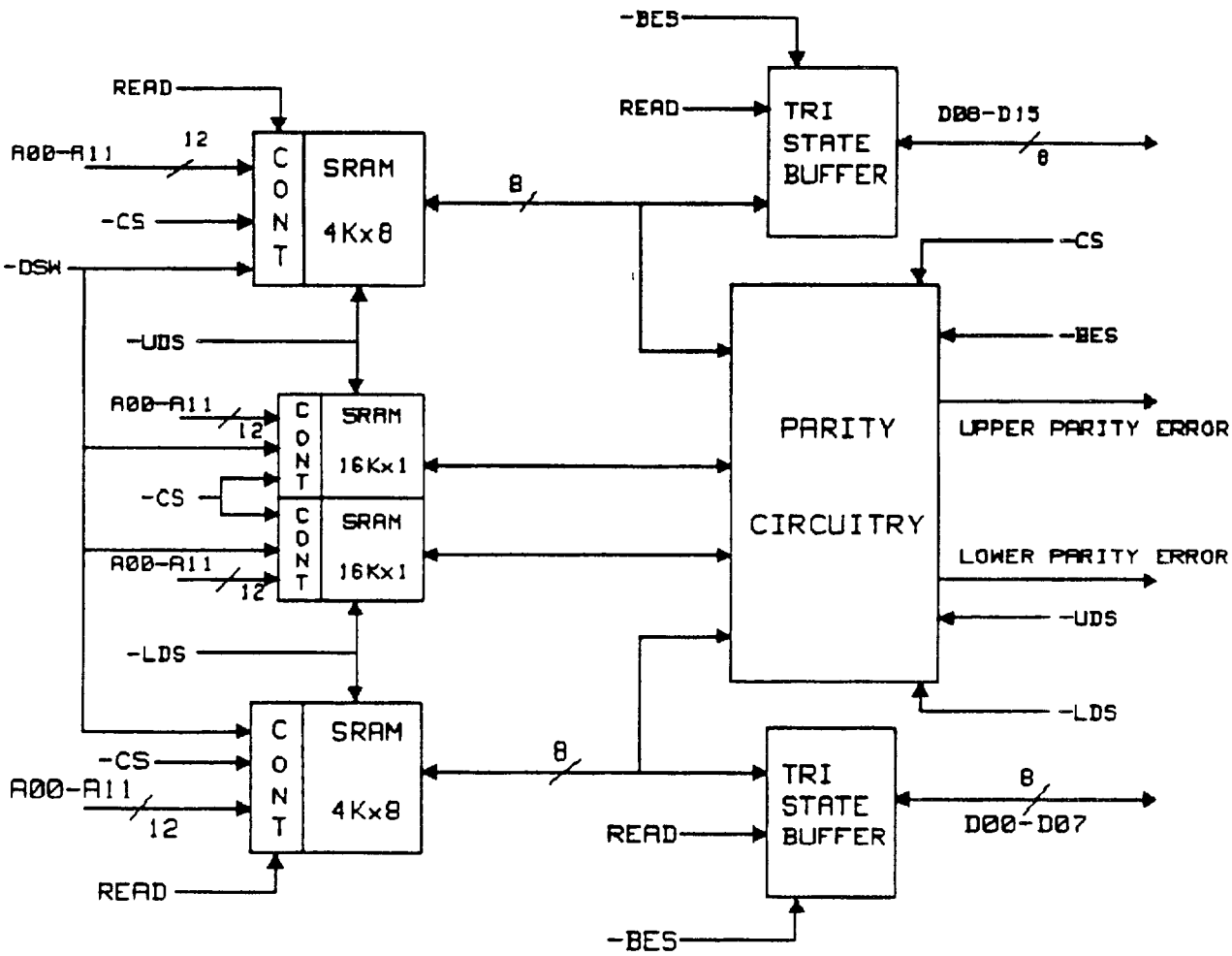


FIGURE 1. DGG functional block diagram.

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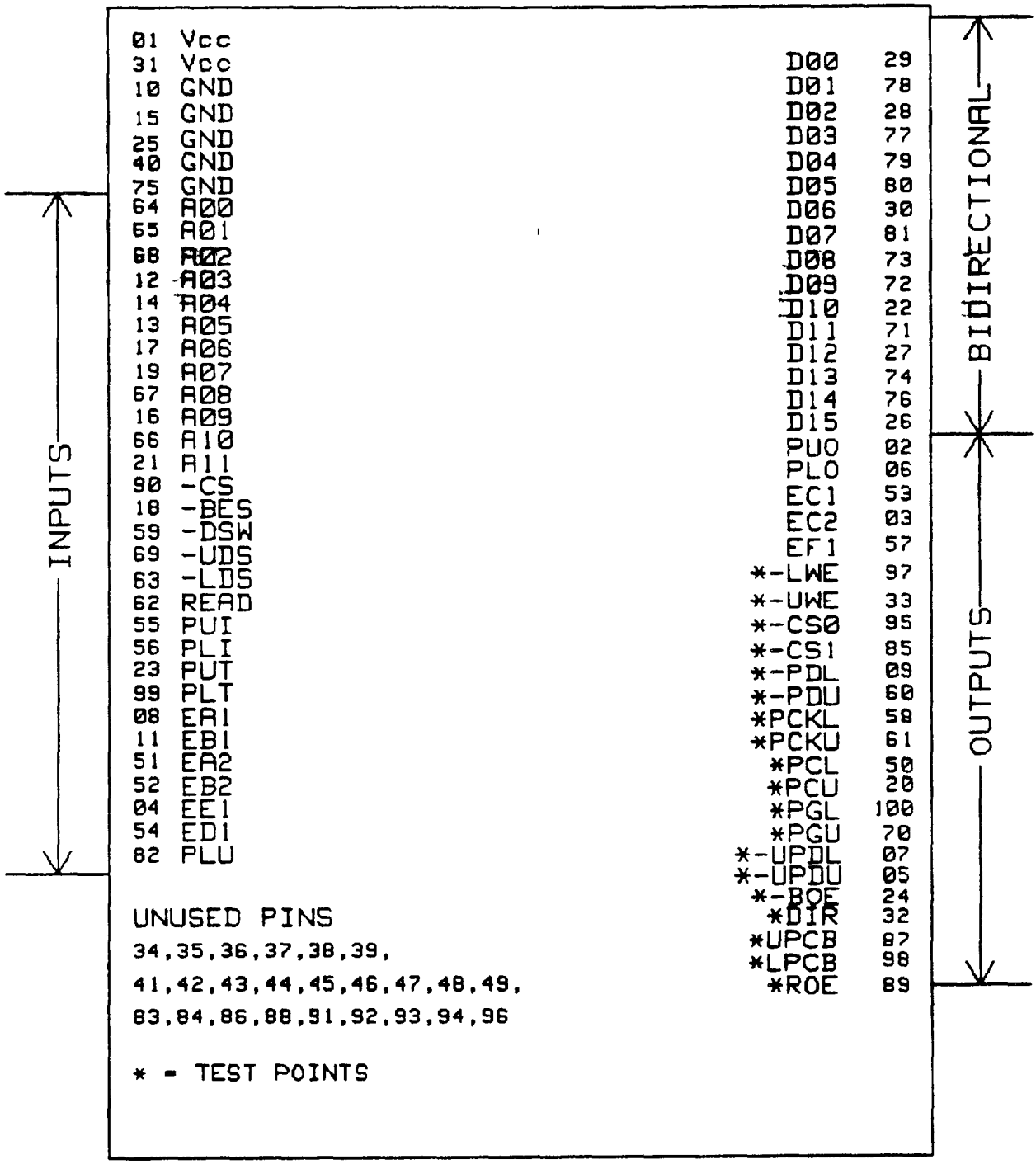


FIGURE 2. DGG pin assignment.

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4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Quality Assurance Activity (SEM-QAA) and they conform to the requirements of MIL-M-28787.

4.4.2 Order of testing. Test conditions A through Y shall be performed in any order with the following exceptions. Test conditions A, C, D, E and F are stress tests and shall be performed only at 25 ± 5°C. If test conditions A, C, D, E, or F are part of a sequence of tests, the order of testing shall be as follows: A, C, D, E and F where applicable, followed by the remaining tests in the sequence.

4.4.3 Multiple testing of pins. Singular pins listed multiple times in a test table indicate multiple devices connected to the pins. The module shall be tested so that each device connected to a pin is exercised and tested.

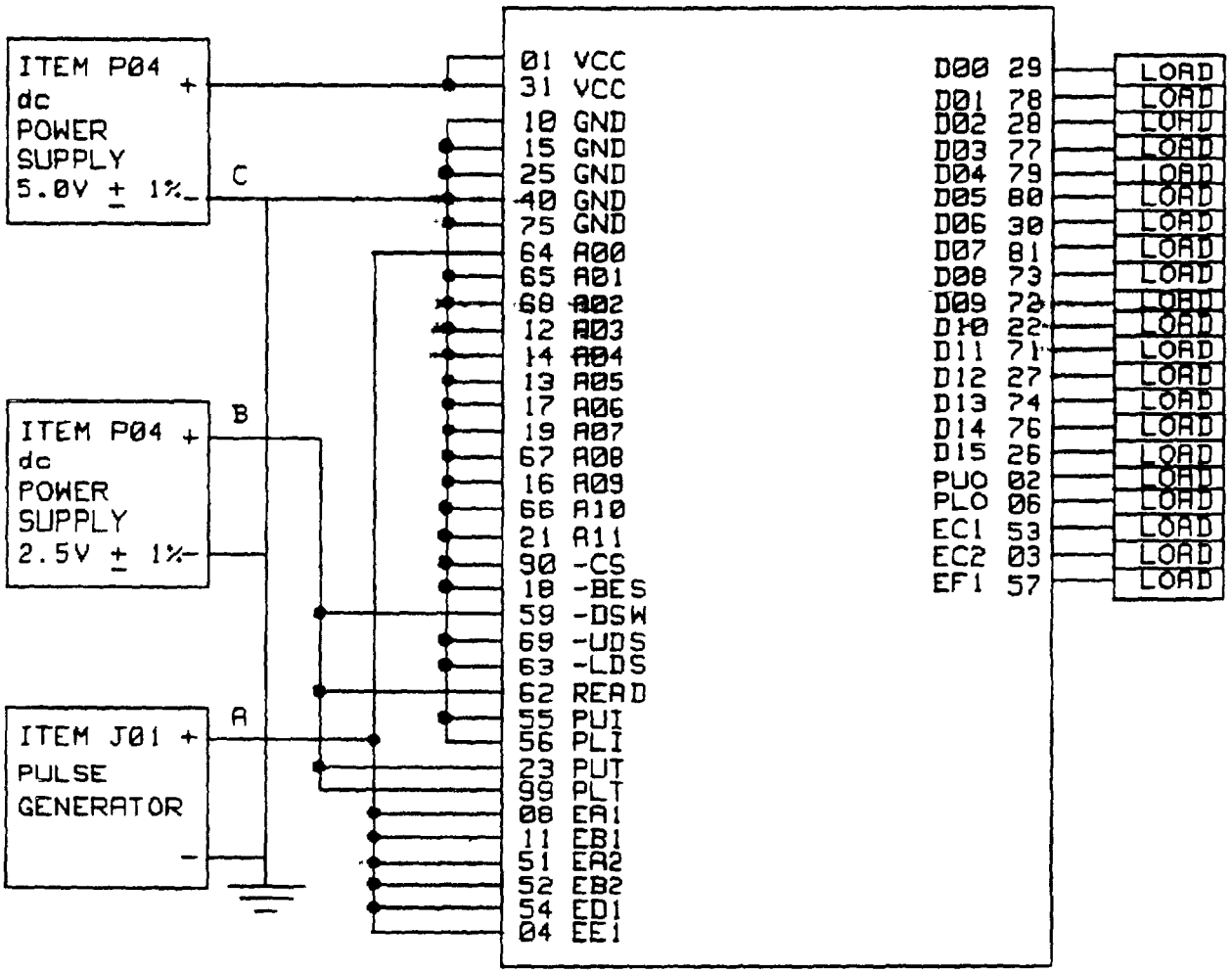
4.5 Qualification testing. Initial qualification tests and periodic check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through Y in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions B, D through V, X and Y shall be performed in the order specified in 4.4.2 using the initial limits for the applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 3. The input waveforms are shown on figure 4.

4.5.1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figures 5 and 6.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 7. Following the vibration tests, the electrical tests in table I which are marked with an "X" in the test type column, and test condition R, shall be performed using the initial limits at 25 ± 5°C ambient.

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NOTE:

1. The load for each output is as specified in the output voltage test.

FIGURE 3. DGG operating temperature and life test circuit.

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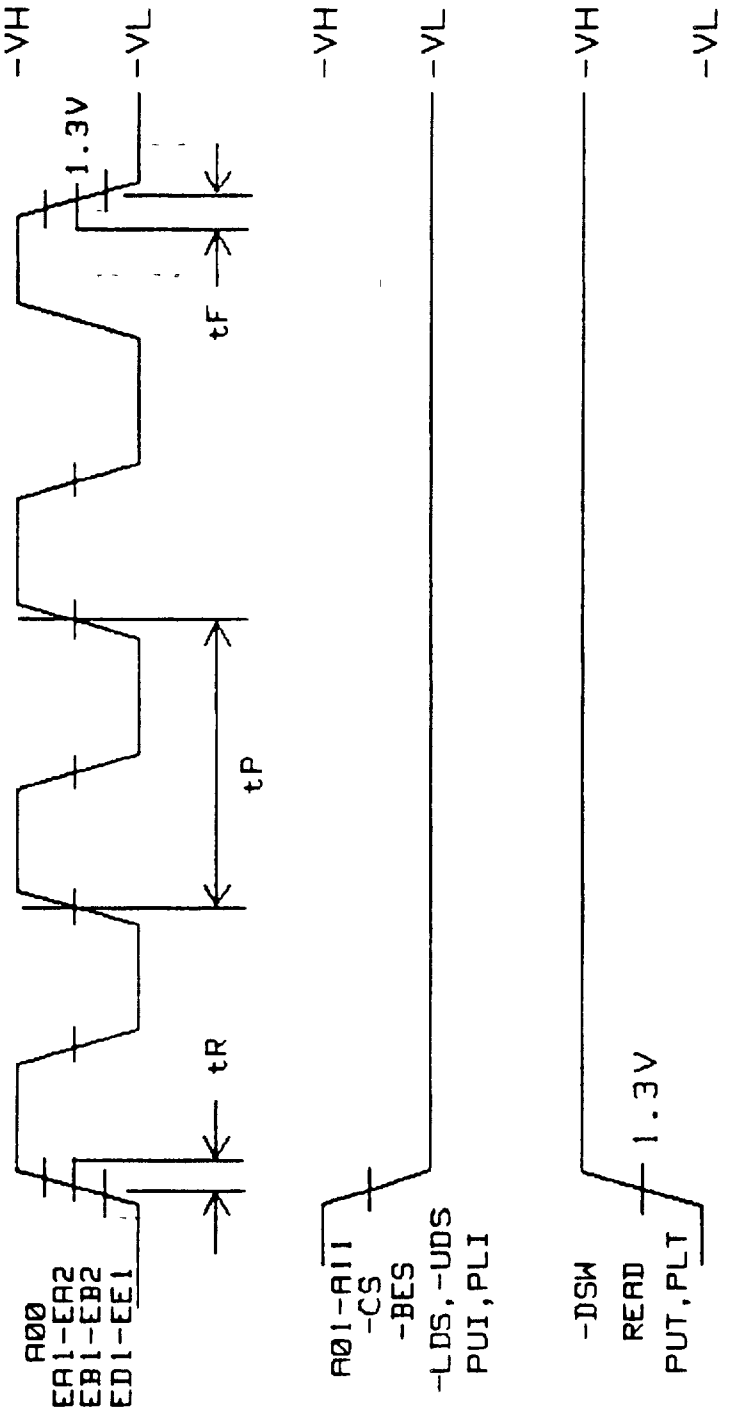


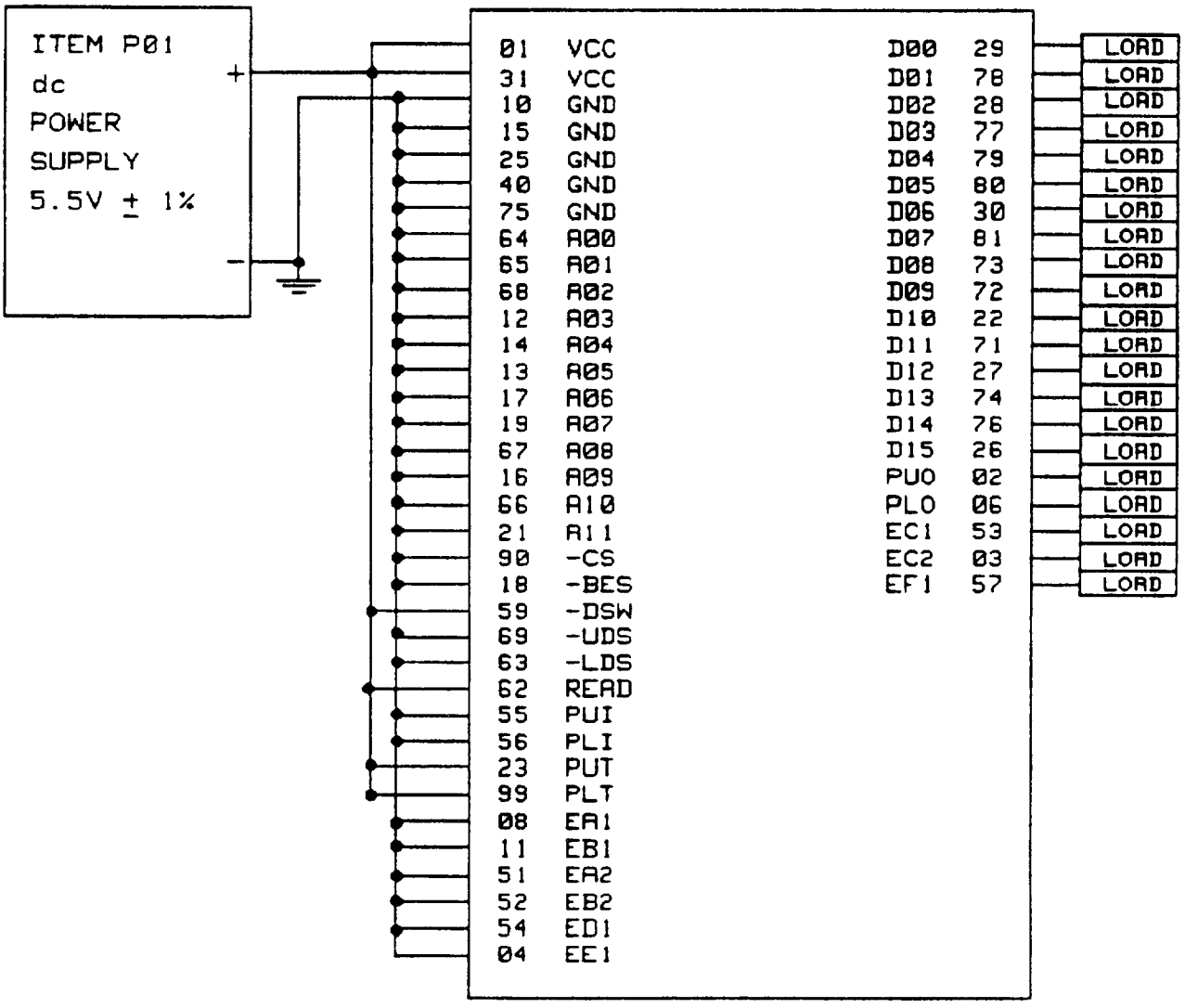
FIGURE 4. Operating temperature and life test waveforms.

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NOTES:

1. tr = 15 ± 1 ns.
2. tf = 6 ± 1 ns.
3. tp = 400 ± 10 ns.
4. Vref = 1.3 V.
5. VH = 3.0 ± 0.1 V.
6. VL = 0.0 ± 0.2 V.

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NOTE:

1. The load for each output is as specified in the output voltage test.

FIGURE 5. DGG component temperature test circuit.

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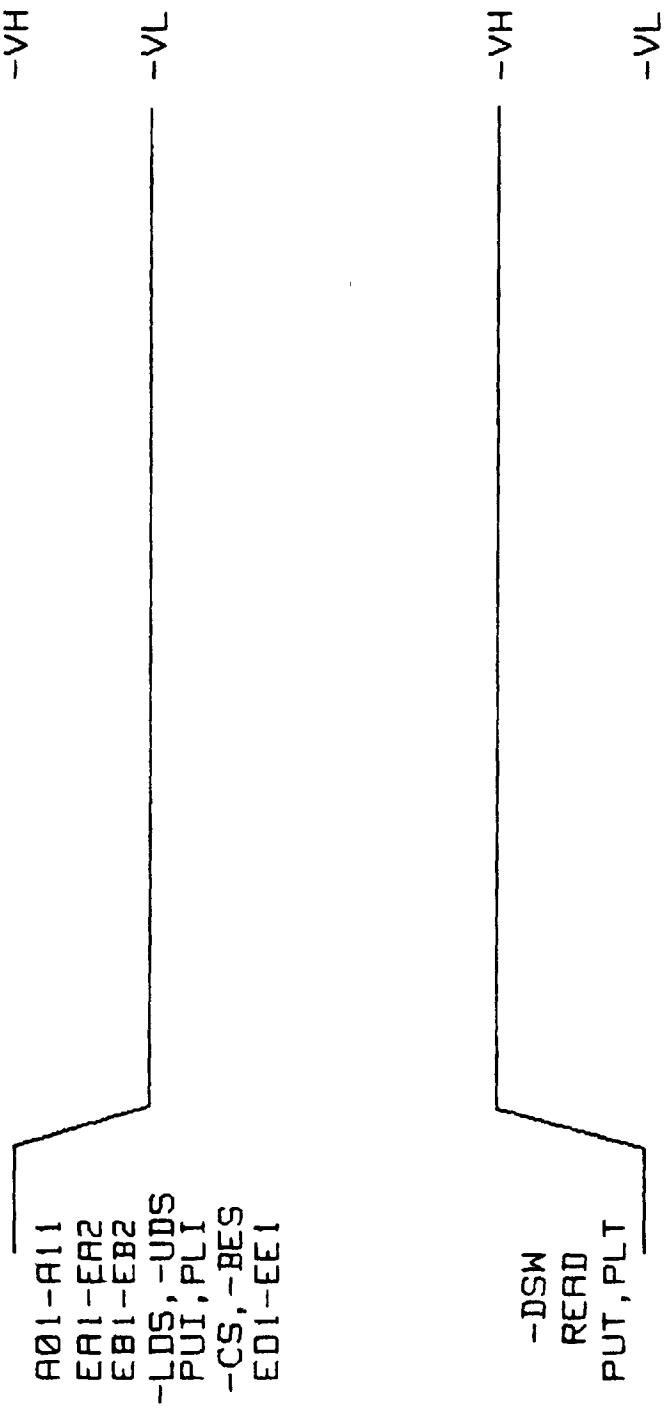


FIGURE 6. Component temperature test waveforms.

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NOTES:

1. VH = 5.5 V ± 1.0 percent.

2. VL = 0.0 ± 0.1 V.

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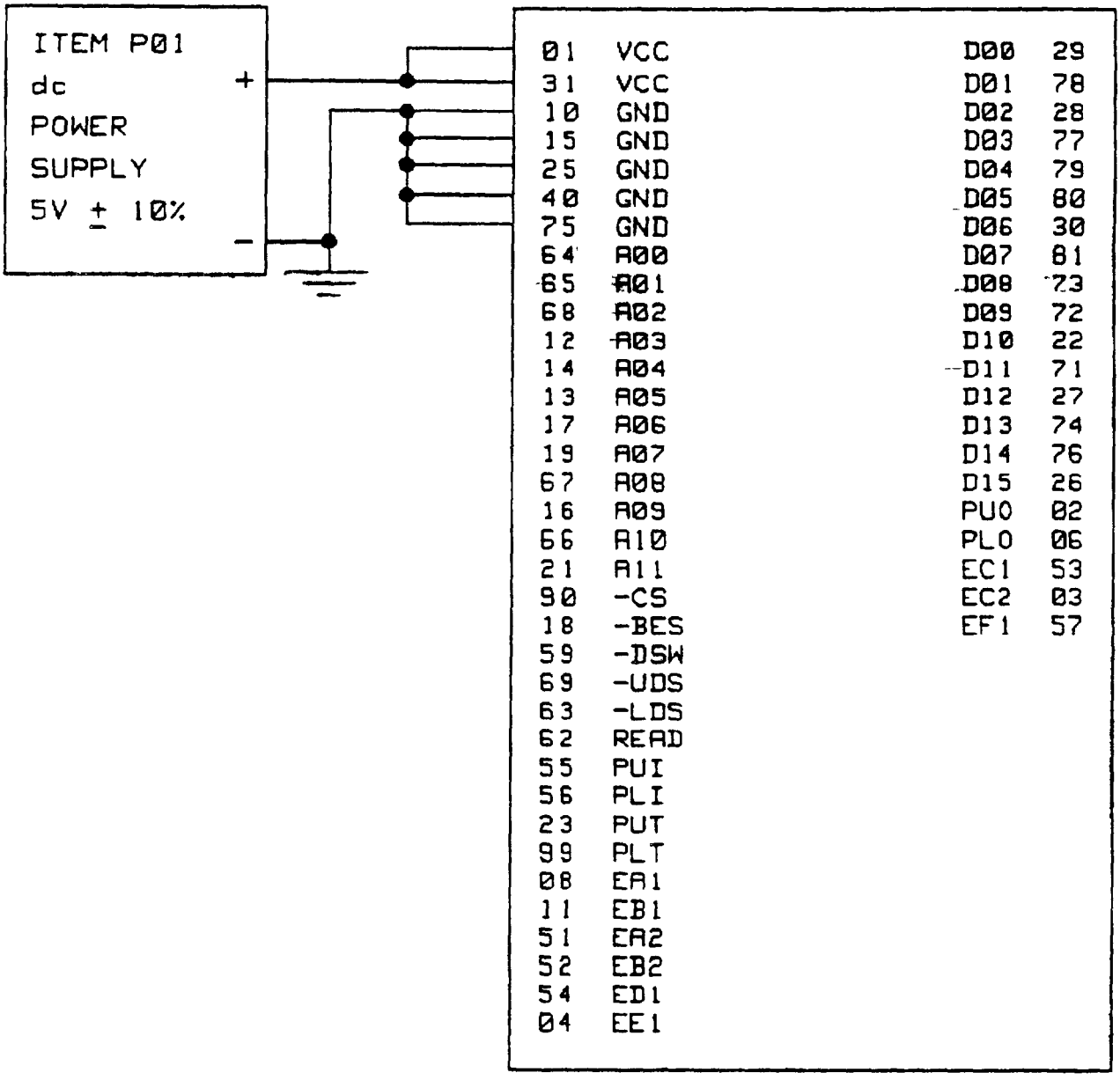


FIGURE 7. DGG vibration test circuit.

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4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figures 3 and 4. Following the life test and while still at the maximum class temperature, electrical tests B, D through U, X, and Y shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements in table I. Following the return of the module to 25 ± 5°C, electrical tests A1, B, D through U, X, Y shall be performed in the order specified in 4.4.2 using the 25 ± 5°C ambient end-of-life requirements in table I.

4.5.4 Thermal shock test. The thermal shock test shall be conducted in accordance with MIL-M-28787. The module shall be unpowered during this test. Following the thermal shock test, the electrical tests in table I which are marked with an "X" in the test type column, shall be performed using the initial limits at 25 ± 5°C ambient.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Definitions. Negative (-) current is current directed out of the module pin and positive (+) current is current directed into the module pin. Abbreviations used in this specification are defined as follows:

(a) Cin = input capacitance.
(b) Cout = output capacitance.
(c) Ibd = input breakdown current.
(d) Icc = power supply current.
(e) Iih = input high level current.
(f) Iil = input low level current.
(g) Iin = input forcing current.
(h) Ioh = output high level current.
(i) Iol = output low level current.
(j) Ios = output short circuit current.
(k) Iozh = output high level leakage current.
(l) Iozl = output low level leakage current.
(m) NA = not applicable.
(n) PSFC = power supply filter capacitance.
(o) PSTA = power supply transient amplitude.
(p) Vic = input clamp diode voltage.
(q) Vih = input high level voltage.
(r) Vil = input low level voltage.
(s) Vin = input forcing voltage.
(t) Voh = output high level voltage.
(u) Vol = output low level voltage.
(v) Vout = output forcing voltage.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements in table I, the module is considered to have completed its useful life (see 3.9).

6.3 Module description. The following electrical parameters reflect the 0°C to 60°C end-of-life parameters in table I. Operating power supply voltage range: 4.5 V dc to 5.5 V dc. Module power dissipation: 4.502 W (max); 2.334 W (typical).

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6.3.1 Input parameters.

Signal	Vic min (V)	Vil max (V)	Vih min (V)	Iil min (mA)	Iih max (uA)	Ibd max (uA)	Cin max (pF)
A00	-1.5	0.7	2.0	-0.4	20.0	100	30.0
A01	-1.5	0.7	2.0	-0.4	20.0	100	30.0
A02	-1.5	0.7	2.0	-0.4	20.0	100	30.0
A03	-1.5	0.7	2.0	-0.4	20.0	100	30.0
A04	-1.5	0.7	2.0	-0.4	20.0	100	30.0
A05	-1.5	0.7	2.0	-0.4	20.0	100	30.0
A06	-1.5	0.7	2.0	-0.4	20.0	100	30.0
A07	-1.5	0.7	2.0	-0.4	20.0	100	30.0
A08	-1.5	0.7	2.0	-0.4	20.0	100	30.0
A09	-1.5	0.7	2.0	-0.4	20.0	100	30.0
A10	-1.5	0.7	2.0	-0.4	20.0	100	30.0
A11	-1.5	0.7	2.0	-0.25	25.0	1000	30.0
D00	-1.5	0.7	2.0	-0.2	20.0	100	60.0
D01	-1.5	0.7	2.0	-0.2	20.0	100	60.0

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6.3.1 Input parameters - Continued.

Signal	Vic min (V)	Vil max (V)	Vih min (V)	Iil min (mA)	Iih max (uA)	Ibd max (uA)	Cin max (pF)
D02	-1.5	0.7	2.0	-0.2	20.0	100	60.0
D03	-1.5	0.7	2.0	-0.2	20.0	100	60.0
D04	-1.5	0.7	2.0	-0.2	20.0	100	60.0
D05	-1.5	0.7	2.0	-0.2	20.0	100	60.0
D06	-1.5	0.7	2.0	-0.2	20.0	100	60.0
D07	-1.5	0.7	2.0	-0.2	20.0	100	60.0
D08	-1.5	0.7	2.0	-0.2	20.0	100	60.0
D09	-1.5	0.7	2.0	-0.2	20.0	100	60.0
D10	-1.5	0.7	2.0	-0.2	20.0	100	60.0
D11	-1.5	0.7	2.0	-0.2	20.0	100	60.0
D12	-1.5	0.7	2.0	-0.2	20.0	100	60.0
D13	-1.5	0.7	2.0	-0.2	20.0	100	60.0
D14	-1.5	0.7	2.0	-0.2	20.0	100	60.0
D15	-1.5	0.7	2.0	-0.2	20.0	100	60.0

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6.3.1 Input parameters - Continued.

Signal	Vic min (V)	Vil max (V)	Vih min (V)	Iil min (mA)	Iih max (uA)	Ibd max (uA)	Cin max (pF)
-CS	-1.5	0.7	2.0	-2.0	50.0	1000	60.0
-BES	-1.2	0.7	2.0	-2.4	70.0	1100	60.0
-DSW	-1.5	0.7	2.0	-0.8	40.0	200	60.0
-UDS	-1.5	0.7	2.0	-0.65	45.0	1100	60.0
-LDS	-1.5	0.7	2.0	-0.65	45.0	1100	60.0
READ	-1.5	0.7	2.0	-0.65	45.0	1100	60.0
PUI	-1.5	0.7	2.0	-0.488	-227.1	111	60.0
PLI	-1.5	0.7	2.0	-0.488	-227.1	111	60.0
PUT	-1.5	0.7	2.0	-0.37	20.0	100	60.0
PLT	-1.5	0.7	2.0	-0.37	20.0	100	60.0
EA1	-1.5	0.7	2.0	-0.4	20.0	100	30.0
EB1	-1.5	0.7	2.0	-0.4	20.0	100	30.0
EA2	-1.5	0.7	2.0	-0.4	20.0	100	30.0

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6.3.1 Input parameters - Continued.

Signal	Vic min (V)	Vil max (V)	Vih min (V)	Iil min (mA)	Iih max (uA)	Ibd max (uA)	Cin max (pF)
EB2	-1.5	0.7	2.0	-0.4	20.0	100	30.0
ED1	-1.5	0.7	2.0	-0.4	20.0	100	30.0
EE1	-1.5	0.7	2.0	-0.4	20.0	100	30.0
PLU	-1.5	0.7	2.0	-9.43	-2.35	1071	60.0

6.3.2 Output parameters.

Signal	Voh min (V)	Vol max (V)	Ioh min (uA)	Iol max (mA)	Iozh max (uA)	Iozl min (uA)	Cout max (pF)	Ios max (mA)
D00	2.4	0.4	-3000	12	20	-200	60	-40.0
D01	2.4	0.4	-3000	12	20	-200	60	-40.0
D02	2.4	0.4	-3000	12	20	-200	60	-40.0
D03	2.4	0.4	-3000	12	20	-200	60	-40.0
D04	2.4	0.4	-3000	12	20	-200	60	-40.0
D05	2.4	0.4	-3000	12	20	-200	60	-40.0
D06	2.4	0.4	-3000	12	20	-200	60	-40.0

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6.3.2 Output parameters - Continued.

Signal	Voh min (V)	Vol max (V)	Ioh min (uA)	Iol max (mA)	Iozh max (uA)	Iozl min (uA)	Cout max (pF)	Ios max (mA)
D07	2.4	0.4	-3000	12	20	200	60	-40.0
D08	2.4	0.4	-3000	12	20	200	60	-40.0
D09	2.4	0.4	-3000	12	20	200	60	-40.0
D10	2.4	0.4	-3000	12	20	200	60	-40.0
D11	2.4	0.4	-3000	12	20	200	60	-40.0
D12	2.4	0.4	-3000	12	20	200	60	-40.0
D13	2.4	0.4	-3000	12	20	200	60	-40.0
D14	2.4	0.4	-3000	12	20	200	60	-40.0
D15	2.4	0.4	-3000	12	20	200	60	-40.0
PUO	2.5	0.4	-400	4	NA	NA	30	-15.0
PLO	2.5	0.4	-400	4	NA	NA	30	-15.0
EC1	2.5	0.4	-400	4	NA	NA	30	-15.0
EC2	2.5	0.4	-400	4	NA	NA	30	-15.0
EF1	2.5	0.4	-400	4	NA	NA	30	-15.0

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6.3.2 Output parameters - Continued.

Signal	Voh min (V)	Vol max (V)	Ioh min (uA)	Iol max (mA)	Iozh max (uA)	Iozl min (uA)	Cout max (pF)	Ios max (mA)
CS0	2.4	0.5	-980	19.98	NA	NA	60	-40.0
CS1	2.4	0.5	-980	19.98	NA	NA	60	-40.0
UPDL	2.4	0.4	-380	3.6	NA	NA	60	-15.03
UPDU	2.4	0.4	-380	3.6	NA	NA	60	-15.03
PDL	2.4	0.5	-1980	7.6	NA	NA	60	-30.03
PDU	2.4	0.5	-1980	7.6	NA	NA	60	-30.03
PCKL	2.4	0.5	-1960	7.6	NA	NA	60	-30.06
PCKU	2.4	0.5	-1960	7.6	NA	NA	60	-30.06
LPCB	2.4	0.4	-4000	7.75	NA	NA	60	NA
UPCB	2.4	0.4	-4000	7.75	NA	NA	60	NA
PCL	2.4	0.4	-390	3.75	NA	NA	60	-15.0
PCU	2.4	0.4	-390	3.75	NA	NA	60	-15.0
PGL	2.5	0.4	-375	3.75	NA	NA	60	-15.0
PGU	2.5	0.4	-375	3.75	NA	NA	60	-15.0

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6.3.2 Output parameters - Continued.

Signal	Voh min (V)	Vol max (V)	Ioh min (uA)	Iol max (mA)	Iozh max (uA)	Iozl min (uA)	Cout min (pF)	Ios max (mA)
UWE	2.5	0.4	-370	3.98	NA	NA	60	-15.0
LWE	2.5	0.4	-370	3.98	NA	NA	60	-15.0
BOE	2.5	0.4	-360	3.6	NA	NA	60	-15.0
DIR	2.5	0.4	-310	1.6	NA	NA	60	-15.0
ROE	2.5	0.5	-960	19.96	NA	NA	60	-40.0

6.3.3 Dynamic parameters. The dynamic parameters appear with the dynamic test.

6.4 Module function. The DGG module provides 64K bits of RAM storage. Data may be written into the module in either word or byte format. Data must be read in word format. The memory elements are arranged in a 4K X 16 bit pattern. The module also provides parity generation and checking circuitry, which verifies the reliability of the stored data. A logic low on the error output signals that an error exists between the stored and read data. The memory function is activated by addressing memory locations via the address lines A00-A11, and by exercising the six (6) control lines: -CS, -BS, -DSW, -UDS, -LDS and READ. Data are either written to or read from the module by using data bus lines D00-D15. Three spare gates are provided on the module, two (2) AND gates and one (1) OR gate. The inputs are labeled EA1, EB1 and EA2, EB2 for the AND gates, and ED1, EE1 for the OR gate. The outputs are labeled EC1 and EC2 for the AND gates, and EF1 for the OR gate. These gates may be used at the designer's discretion.

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6.4.1 Functional description of I/O pins.

Signal name	Pin number	I/O	Functional description
A00	64	I	Address Lines - used to address memory locations in the module. Address A11 is used to select the upper or lower byte of memory.
A01	65	I	
A02	68	I	
A03	12	I	
A04	14	I	
A05	13	I	
A06	17	I	
A07	19	I	
A08	67	I	
A09	16	I	
A10	66	I	
A11	21	I	
D00	29	I/O	Data Bus - used to transfer data in and out of the module.
D01	78	I/O	
D02	28	I/O	
D03	77	I/O	
D04	79	I/O	
D05	80	I/O	
D06	30	I/O	
D07	81	I/O	
D08	73	I/O	
D09	72	I/O	
D10	22	I/O	
D11	71	I/O	
D12	27	I/O	
D13	74	I/O	
D14	76	I/O	
D15	26	I/O	
-CS	90	I	Module Select - used to select the module (active low).
-BES	18	I	Bus Enable Strobe - used to set up the internal data bus for data transfer and output (active low).
-DSW	59	I	Data Strobe Write - used to strobe data into memory on write cycle (active low).

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6.4.1 Functional description of I/O pins - Continued.

Signal name	Pin number	I/O	Functional description
-UDS -LDS	69 63	I I	Upper, Lower Data Strobe - used to select upper or lower byte for memory write (active low).
READ	62	I	Read - used to set up the direction of the data bus. Also used to clear or set the parity data.
PUI PLI	55 56	I I	Upper, Lower Parity Buffer Input - used to daisy-chain parity information.
PUT PLT	23 99	I I	Upper, Lower Parity Error Generate - used to generate parity error information. May be tied either high or low.
EA1 EB1 EA2 EB2	8 11 51 52	I I I I	Inputs to pinned out AND-GATE - these are spares on the module.
ED1 EE1	54 4	I I	Inputs to pinned out OR-GATE - this is a spare OR-gate on the module.
EC1 EC2 EF1	53 3 57	O O O	Outputs to spare gates.
PLO PUO	6 2	O O	Upper, Lower Buffered Parity Data - these signals are buffered versions of parity error signals.
PLU	82	I	Pull-Up - used to verify pull-up value.

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6.4.1 Functional description of I/O pins - Continued.

Signal name	Pin number	I/O	Functional description
-BOE	24	O	Test Points - to monitor desired functions.
DIR	32	O	
-CS0	95	O	
-CS1	85	O	
-UWE	33	O	
-LWE	97	O	
ROE	89	O	
-PDL	9	O	
-PDU	60	O	
PCKU	61	O	
PCKL	58	O	
PCU	20	O	
PCL	50	O	
PGU	70	O	
-UPDU	5	O	
-UPDL	7	O	
UPCB	87	O	
LPCB	98	O	
PGL	100	O	

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6.5 Description of tests. The following tables and associated figures describe the tests to be performed.

(a) Isolation test - table II.

(b) Continuity test - table III.

(c) Vcc stress test - table IV.

(d) Ibd test - table V.

(e) Vic test - table VI.

(f) Ios test - table VII.

(g) Iih test - table VIII.

(h) Iil test - table IX.

(i) Voh test - table X, figure 8.

(j) Vol test - table XI, figure 8.

(k) Iozh test - table XII.

(l) Iozl test - table XIII.

(m) Icc test - table XIV.

(n) Functional test - table XV, figure 9.

(o) Dynamic test - tables XVI and XVII, figures 10 and 11.

(p) PSFC test - table XVIII.

(q) PSTA test - table XIX, figure 12.

(r) Output crosstalk tests - tables XX and XXI, figures 13 through 15.

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(s) Cin test - table XXII.

(t) Cout test - table XXIII.

(u) Disable time test - table XXIV, figures 16 and 17.

(v) Threshold test - table XXV.

6.6 Functional test vectors. Functional test vectors are not printed herein because of their extreme length; and, even when presented as a hard copy, they require conversion to on-line format. When the entire sequence of vector groups is needed for driving devices as part of inspection, a test program output should be generated in a form compatible with test system architecture (magnetic tape format usually preferred). For reference purposes, the Naval Weapons Support Center (Code 607) maintains a limited quantity of dated, hard copy printouts of the functional test vectors and these may be obtained upon request.

Preparing activity:
Navy - SH

(Project 5963-N746)

Agent: NW

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TABLE II. Isolation test.

Test cond	Test description	Connect A to pin	Connect B to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
A1	Isolation, frame	All pins	Frame	10	INF	MOhm	1-7
A2	Isolation, used to unused	All used pins	All unused pins	10	INF	MOhm	1-7
A3	Isolation, unused to unused	34	35,36,37,38, 39,41,42,43, 44,45,46,47, 48,49,83,84, 86,88,91,92, 93,94,96	10	INF	MOhm	1-7
A4	Isolation, unused to unused	35	36,37,38,39, 41,42,43,44, 45,46,47,48, 49,83,84,86, 88,91,92,93, 94,96				
A5	Isolation, unused to unused	36	37,38,39,41, 42,43,44,45, 46,47,48,49, 83,84,86,88, 91,92,93,94, 96				
A6	Isolation, unused to unused	37	38,39,41,42, 43,44,45,46, 47,48,49,83, 84,86,88,91, 92,93,94,96				
A7	Isolation, unused to unused	38	39,41,42,43, 44,45,46,47, 48,49,83,84, 86,88,91,92, 93,94,96	10	INF	MOhm	1-7

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TABLE II. Isolation test - Continued.

Test cond	Test description	Connect A to pin	Connect B to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
A8	Isolation, unused to unused	39	41,42,43,44, 45,46,47,48, 49,83,84,86, 88,91,92,93, 94,96	10	INF	MOhm	1-7
A9	Isolation, unused to unused	41	42,43,44,45, 46,47,48,49, 83,84,86,88, 91,92,93,94, 96				
A10	Isolation, unused to unused	42	43,44,45,46, 47,48,49,83, 84,86,88,91, 92,93,94,96				
A11	Isolation, unused to unused	43	44,45,46,47, 48,49,83,84, 86,88,91,92, 93,94,96				
A12	Isolation, unused to unused	44	45,46,47,48, 49,83,84,86, 88,91,92,93, 94,96				
A13	Isolation, unused to unused	45	46,47,48,49, 83,84,86,88, 91,92,93,94, 96				
A14	Isolation, unused to unused	46	47,48,49,83, 84,86,88,91, 92,93,94,96				
A15	Isolation, unused to unused	47	48,49,83,84, 86,88,91,92, 93,94,96	10	INF	MOhm	1-7

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TABLE II. Isolation test - Continued.

Test cond	Test description	Connect A to pin	Connect B to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
A16	Isolation, unused to unused	48	49,83,84,86, 88,91,92,93, 94,96	10	INF	MOhm	1-7
A17	Isolation, unused to unused	49	83,84,86,88, 91,92,93,94, 96				
A18	Isolation, unused to unused	83	84,86,88,91, 92,93,94,96				
A19	Isolation, unused to unused	84	86,88,91,92, 93,94,96				
A20	Isolation, unused to unused	86	88,91,92,93, 94,96				
A21	Isolation, unused to unused	88	91,92,93,94, 96				
A22	Isolation, unused to unused	91	92,93,94,96				
A23	Isolation, unused to unused	92	93,94,96				
A24	Isolation, unused to unused	93	94,96				
A25	Isolation, unused to unused	94	96	10	INF	Mohm	1-7

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NOTES:

1. Isolation test tolerance table.

Condition	Value	Accuracy	Test condition
Megohm bridge voltage	100 V 30 V	± 10 percent ± 10 percent	A1,A2 A3-A25
Isolation measurement (see note 6)	NA NA	± 500 KOhm ± 50 KOhm	A1-A25 A1-A25

2. The ohmmeter used shall be item R06.

3. In the setup conditions in table II, 'A' shall correspond to the positive terminal of the megohm bridge, and 'B' shall correspond to the negative terminal.

4. Warning: When using a megohm bridge, high voltage may be present at any of the output binding posts. Always set the megohm bridge function switch to "DISCHARGE" before connecting or disconnecting the test leads.

5. When testing for frame isolation, penetrate the anodized finish on the module frame with a sharply pointed probe connected to the negative terminal of the megohm bridge.

6. The minimum end-of-life limit is 1 MOhm.

7. Abbreviations used are defined in 6.1.

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TABLE III. Continuity test.

Test cond	Test description	Connect A to pin	Connect B to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
B1	Continuity, GND	10	15	0.0	1.0	Ohm	1-4
B2	Continuity, GND	15	25				
B3	Continuity, GND	25	40				
B4	Continuity, GND	40	75				
B5	Continuity, Vcc	1	31	0.0	1.0	Ohm	1-4
B6	Continuity, GND	10	15	0.0	0.1	Ohm	1-4
B7	Continuity, GND	15	25				
B8	Continuity, GND	25	40				
B9	Continuity, GND	40	75				
B10	Continuity, Vcc	1	31	0.0	0.1	Ohm	1-4

NOTES:

1. Continuity test tolerance table.

Condition	Value	Accuracy	Test condition
Continuity measurement	NA NA	± 50 mOhm ± 5 mOhm	B1-B5 B6-B10

2. The ohmmeter used shall be item R07.

3. In the setup conditions in table III, 'A' shall correspond to the positive terminal of the ohmmeter, and 'B' shall correspond to the negative terminal.

4. Abbreviations used are defined in 6.1.

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TABLE IV. Power supply stress test.

Test cond	Power supply	Vin (V)	Connect Positive lead to	Connect Negative lead to	25 ± 5°C Initial limits	Notes
C	Vcc	7.0	1,31	10,15 25,40, 75	Module shall pass all remaining tests.	1-5

NOTES:

1. Power supply stress test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	7.0 V	+0 mV,-70 mV	C
Power supply current limit	1.275 A	NA	C

2. There shall be no power supply voltage overshoot on this test.

3. Connect the power supplies to the module as indicated in table IV.

4. When performing the power supply stress test, no measurements are taken. This condition shall be maintained for a minimum of 5 seconds.

5. Abbreviations used are defined in 6.1.

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TABLE V. Input breakdown current test.

Test cond	Signal name	Pin no.	Vin (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
D1	A00	64	5.5	0.0	0.1	mA	1-3
D2	A01	65					
D3	A02	68					
D4	A03	12					
D5	A04	14					
D6	A05	13					
D7	A06	17					
D8	A07	19					
D9	A08	67					
D10	A09	16					
D11	A10	66	5.5	0.0	0.1	mA	1-3
D12	A11	21	5.5	0.0	1.0	mA	1-3
D13	D00	29	5.5	0.0	0.1	mA	1-3
D14	D01	78					
D15	D02	28					
D16	D03	77					
D17	D04	79					
D18	D05	80					
D19	D06	30					
D20	D07	81	5.5	0.0	0.1	mA	1-3

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TABLE V. Input breakdown current test - Continued.

Test cond	Signal name	Pin no.	Vin (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
D21	D08	73	5.5	0.0	0.1	mA	1-3
D22	D09	72					
D23	D10	22					
D24	D11	71					
D25	D12	27					
D26	D13	74					
D27	D14	76					
D28	D15	26	5.5	0.0	0.1	mA	1-3
D29	-CS	90	5.5	0.0	1.0	mA	1-3
D30	-BES	18	5.5	0.0	1.1	mA	1-3
D31	-DSW	59	5.5	0.0	0.2	mA	1-3
D32	-UDS	69	5.5	0.0	1.1	mA	1-3
D33	-LDS	63					
D34	READ	62	5.5	0.0	1.1	mA	1-3
D35	PUI	55	5.5	-0.01	0.111	mA	1-3
D36	PLI	56	5.5	-0.01	0.111	mA	1-3
D37	PUT	23	5.5	0.0	0.1	mA	1-3
D38	PLT	99					
D39	EA1	8					
D40	EB1	11	5.5	0.0	0.1	mA	1-3

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TABLE V. Input breakdown current test - Continued.

Test cond	Signal name	Pin no.	Vin (V)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
D41	EA2	51	5.5	0.0	0.1	mA	1-3
D42	EB2	52					
D43	ED1	54					
D44	EE1	4	5.5	0.0	0.1	mA	1-3
D45	PLU	82	5.5	-0.1	1.071	mA	1-3

NOTES:

1. Input breakdown current test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	D1-D45
Power supply current limit	1.275 A	NA	D1-D45
Forcing voltage, (Vin)	5.5 V 5.5 V 5.5 V	± 55 mV ± 0 mV,-55 mV ± 55 mV	D1-D34 D35,D36 D37-D45
Pattern voltage high, (VH) low, (VL)	3.0 V 0.0 V	± 100 mV ± 100 mV	D1-D45 D1-D45

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NOTES: - Continued.

Condition	Value	Accuracy	Test condition
Input current measurement	NA	± 5.0 uA	D1-D11
	NA	± 50.0 uA	D12
	NA	± 5.0 uA	D13-D28
	NA	± 50.0 uA	D29
	NA	± 55.0 uA	D30
	NA	± 10.0 uA	D31
	NA	± 55.0 uA	D32-D34
	NA	± 6.05 uA	D35,D36
	NA	± 5.0 uA	D37-D44
	NA	± 58.5 uA	D45

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table V and the current then measured. The forcing voltage overshoot shall not exceed 100 mV.

3. Abbreviations used are defined in 6.1.

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TABLE VI. Input clamp diode voltage test.

Test cond	Signal name	Pin no.	Iin (mA)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
E1	A00	64	-18.0	-1.5	-0.4	V	1-3
E2	A01	65					
E3	A02	68					
E4	A03	12					
E5	A04	14					
E6	A05	13					
E7	A06	17					
E8	A07	19					
E9	A08	67					
E10	A09	16					
E11	A10	66					
E12	A11	21					
E13	D00	29					
E14	D01	78					
E15	D02	28					
E16	D03	77					
E17	D04	79					
E18	D05	80					
E19	D06	30					
E20	D07	81	-18.0	-1.5	-0.4	V	1-3

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TABLE VI. Input clamp diode voltage test - Continued.

Test cond	Signal name	Pin no.	I_{IN} (mA)	25 ± 5°C Initial limits Min	Max	Units	Notes
E21	D08	73	-18.0	-1.5	-0.4	V	1-3
E22	D09	72					
E23	D10	22					
E24	D11	71					
E25	D12	27					
E26	D13	74					
E27	D14	76					
E28	D15	26					
E29	-CS	90	-18.0	-1.5	-0.4	V	1-3
E30	-BES	18	-18.0	-1.2	-0.4	V	1-3
E31	-DSW	59	-18.0	-1.5	-0.4	V	1-3
E32	-UDS	69					
E33	-LDS	63					
E34	READ	62					
E35	PUT	23					
E36	PLT	99					
E37	EA1	8					
E38	EB1	11					
E39	EA2	51					
E40	EB2	52	-18.0	-1.5	-0.4	V	1-3

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TABLE VI. Input clamp diode voltage test - Continued.

Test cond	Signal name	Pin no.	Iin (mA)	25 ± 5°C Initial limits Min	Max	Units	Notes
E41	ED1	54	-18.0	-1.5	-0.4	V	1-3
E42	EE1	4	-18.0	-1.5	-0.4	V	1-3

NOTES:

1. Input clamp diode voltage test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	4.5 V	± 45 V	E1-E42
Power supply current limit	1.275 A	NA	E1-E42
Forcing current, (Iin)	-18 mA	± 1.8 mA	E1-E42
Pattern voltage high, (VH) low, (VL)	3.0 V 0.0 V	± 100 mV ± 100 mV	E1-E42 E1-E42
Input voltage measurement	NA NA NA	± 55 mV ± 40 mV ± 55 mV	E1-E29 E30 E31-E42

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NOTES: - Continued.

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing current shall be applied to the pin specified in the appropriate test conditions in table VI and the voltage then measured.

3. Abbreviations used are defined in 6.1.

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TABLE VII. Output short circuit current test.

Test cond	Signal name	Pin no.	Vout (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
F1	D00	29	0.0	-225.0	-40.0	mA	1-4
F2	D01	78					
F3	D02	28					
F4	D03	77					
F5	D04	79					
F6	D05	80					
F7	D06	30					
F8	D07	81					
F9	D08	73					
F10	D09	72					
F11	D10	22					
F12	D11	71					
F13	D12	27					
F14	D13	74					
F15	D14	76					
F16	D15	26	0.0	-225.0	-40.0	mA	1-4
F17	PUO	2	0.0	-110.0	-15.0	mA	1-4
F18	PLO	6					
F19	EC1	53					
F20	EC2	3	0.0	-110.0	-15.0	mA	1-4

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TABLE VII. Output short circuit current test - Continued.

Test cond	Signal name	Pin no.	Vout (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
F21	EF1	57	0.0	-110.0	-15.0	mA	1-4
F22	UWE	33	0.0	-110.0	-15.0	mA	1-4
F23	CS0	95	0.0	-100.0	-40.0	mA	1-4
F24	CS1	85	0.0	-100.0	-40.0	mA	1-4
F25	LWE	97	0.0	-110.0	-15.0	mA	1-4
F26	PDL	9	0.0	-130.4	-30.03	mA	1-4
F27	PDU	60	0.0	-130.4	-30.03	mA	1-4
F28	PCKL	58	0.0	-130.8	-30.06	mA	1-4
F29	PCKU	61	0.0	-130.8	-30.06	mA	1-4
F30	PCL	50	0.0	-100.0	-15.0	mA	1-4
F31	PCU	20	0.0	-100.0	-15.0	mA	1-4
F32	PGL	100	0.0	-100.25	-15.0	mA	1-4
F33	PGU	70	0.0	-100.25	-15.0	mA	1-4
F34	UPDL	7	0.0	-100.4	-15.03	mA	1-4
F35	UPDU	5	0.0	-100.4	-15.03	mA	1-4
F36	BOE	24	0.0	-110.4	-15.0	mA	1-4
F37	DIR	32	0.0	-102.4	-15.0	mA	1-4
F38	ROE	89	0.0	-100.0	-40.0	mA	1-4

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NOTES:

1. Output short circuit current test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	F1-F38
Power supply current limit	1.275 A	NA	F1-F38
Forcing voltage, (Vout)	0.0 V	± 20 mV	F1-F38
Pattern voltage			
high, (VH)	3.0 V	± 100 mV	F1-F38
low, (VL)	0.0 V	± 100 mV	F1-F38
Output current measurement	NA	± 9.25 mA	F1-F16
	NA	± 4.75 mA	F17-F22
	NA	± 3.00 mA	F23,F24
	NA	± 4.75 mA	F25
	NA	± 5.02 mA	F26,F27
	NA	± 5.04 mA	F28,F29
	NA	± 4.25 mA	F30,F31
	NA	± 4.26 mA	F32,F33
	NA	± 4.27 mA	F34,F35
	NA	± 4.77 mA	F36
	NA	± 4.37 mA	F37
	NA	± 3.00 mA	F38

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

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NOTES: - Continued.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table VII and the current then measured. The forcing voltage undershoot shall not exceed 100 mV.

3. When performing this test, only one output shall be tested at a time, and the duration of momentarily grounding each output shall not exceed 1 second.

4. Abbreviations used are defined in 6.1.

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TABLE VIII. Input current, high level, test.

Test cond	Signal name	Pin no.	Vih (V)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
G1	A00	64	2.7	0.0	20.0	uA	1-3
G2	A01	65					
G3	A02	68					
G4	A03	12					
G5	A04	14					
G6	A05	13					
G7	A06	17					
G8	A07	19					
G9	A08	67					
G10	A09	16					
G11	A10	66	2.7	0.0	20.0	uA	1-3
G12	A11	21	2.4	0.0	25.0	uA	1-3
G13	D00	29	2.7	0.0	20.0	uA	1-3
G14	D01	78					
G15	D02	28					
G16	D03	77					
G17	D04	79					
G18	D05	80					
G19	D06	30					
G20	D07	81	2.7	0.0	20.0	uA	1-3

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TABLE VIII. Input current, high level, test - Continued.

Test cond	Signal name	Pin no.	Vih (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
G21	D08	73	2.7	0.0	20.0	uA	1-3
G22	D09	72					
G23	D10	22					
G24	D11	71					
G25	D12	27					
G26	D13	74					
G27	D14	76					
G28	D15	26	2.7	0.0	20.0	uA	1-3
G29	-CS	90	2.7	0.0	50.0	uA	1-3
G30	-BES	18	2.7	0.0	70.0	uA	1-3
G31	-DSW	59	2.7	0.0	40.0	uA	1-3
G32	-UDS	69	2.7	0.0	45.0	uA	1-3
G33	-LDS	63					
G34	READ	62	2.7	0.0	45.0	uA	1-3
G35	PUI	55	2.7	-310.0	-200.0	uA	1-3
G36	PLI	56	2.7	-310.0	-200.0	uA	1-3
G37	PUT	23	2.7	0.0	20.0	uA	1-3
G38	PLT	99					
G39	EA1	8					
G40	EB1	11	2.7	0.0	20.0	uA	1-3

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TABLE VIII. Input current, high level, test - Continued.

Test cond	Signal name	Pin no.	Vih (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
G41	EA2	51	2.7	0.0	20.0	uA	1-3
G42	EB2	52					
G43	ED1	54					
G44	EE1	4	2.7	0.0	20.0	uA	1-3
G45	PLU	82	2.7	-3.04	-2.35	mA	1-3

NOTES:

1. Input current, high level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	G1-G45
Power supply current limit	1.275 A	NA	G1-G45
Forcing voltage, (Vih)	2.7 V 2.4 V 2.7 V	± 135 mV ± 120 mV ± 135 mV	G1-G11 G12 G13-G45
Pattern voltage high, (VH) low, (VL)	3.0 V 0.0 V	± 100 mV ± 100 mV	G1-G45 G1-G45

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NOTES: - Continued.

Condition	Value	Accuracy	Test condition
Input current measurement	NA	± 1.0 uA	G1-G11
	NA	± 1.25 uA	G12
	NA	± 1.0 uA	G13-G28
	NA	± 2.5 uA	G29
	NA	± 3.5 uA	G30
	NA	± 2.0 uA	G31
	NA	± 2.25 uA	G32-G34
	NA	± 5.5 uA	G35,G36
	NA	± 1.0 uA	G37-G44
	NA	± 1.52 uA	G45

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table VIII and the current then measured. The forcing voltage overshoot shall not exceed 100 mV.

3. Abbreviations used are defined in 6.1.

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TABLE IX. Input current, low level, test.

Test cond	Signal name	Pin no.	Vil (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
H1	A00	64	0.4	-400.0	-30.0	uA	1-3
H2	A01	65					
H3	A02	68					
H4	A03	12					
H5	A04	14					
H6	A05	13					
H7	A06	17					
H8	A07	19					
H9	A08	67					
H10	A09	16					
H11	A10	66	0.4	-400.0	-30.0	uA	1-3
H12	A11	21	0.4	-250.0	0.0	uA	1-3
H13	D00	29	0.4	-200.0	0.0	uA	1-3
H14	D01	78					
H15	D02	28					
H16	D03	77					
H17	D04	79					
H18	D05	80					
H19	D06	30					
H20	D07	81	0.4	-200.0	0.0	uA	1-3

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TABLE IX. Input current, low level, test - Continued.

Test cond	Signal name	Pin no.	Vil (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
H21	D08	73	0.4	-200.0	0.0	uA	1-3
H22	D09	72					
H23	D10	22					
H24	D11	71					
H25	D12	27					
H26	D13	74					
H27	D14	76					
H28	D15	26	0.4	-200.0	0.0	uA	1-3
H29	-CS	90	0.5	-2.0	0.0	mA	1-3
H30	-BES	18	0.4	-2.4	-0.03	mA	1-3
H31	-DSW	59	0.4	-800.0	-60.0	uA	1-3
H32	-UDS	69	0.4	-650.0	-30.0	uA	1-3
H33	-LDS	63					
H34	READ	62	0.4	-650.0	-30.0	uA	1-3
H35	PUI	55	0.4	-921.1	-488.6	uA	1-3
H36	PLI	56	0.4	-921.1	-488.6	uA	1-3
H37	PUT	23	0.4	-370.0	-120.0	uA	1-3
H38	PLT	99	0.4	-370.0	-120.0	uA	1-3
H39	EA1	8	0.4	-400.0	-30.0	uA	1-3
H40	EB1	11	0.4	-400.0	-30.0	uA	1-3

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TABLE IX. Input current, low level, test - Continued.

Test cond	Signal name	Pin no.	Vil (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
H41	EA2	51	0.4	-400.0	-30.0	uA	1-3
H42	EB2	52					
H43	ED1	54					
H44	EE1	4	0.4	-400.0	-30.0	uA	1-3
H45	PLU	82	0.4	-9.43	-5.0	mA	1-3

NOTES:

1. Input current, low level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	H1-H45
Power supply current limit	1.275 A	NA	H1-H45
Forcing voltage, (Vil)	0.4 V 0.5 V 0.4 V	± 20 mV ± 25 mV ± 20 mV	H1-H28 H29 H30-H45
Pattern voltage high, (VH) low, (VL)	3.0 V 0.0 V	± 100 mV ± 100 mV	H1-H45 H1-H45

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NOTES: - Continued.

Condition	Value	Accuracy	Test condition
Input current measurement	NA	± 18.5 uA	H1-H11
	NA	± 12.5 uA	H12
	NA	± 10.0 uA	H13-H28
	NA	± 100.0 uA	H29
	NA	± 118.0 uA	H30
	NA	± 37.0 uA	H31
	NA	± 31.0 uA	H32-H34
	NA	± 21.6 uA	H35,H36
	NA	± 12.5 uA	H37,H38
	NA	± 18.5 uA	H39-H44
	NA	± 221.0 uA	H45

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table IX and the current then measured. The forcing voltage overshoot shall not exceed 100 mV.

3. Abbreviations used are defined in 6.1.

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TABLE X. Output voltage, high level, test.

Test cond	Signal name	Pin no.	Ioh (mA)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
J1	D00	29	-3.0	2.4	4.5	V	1-4
J2	D01	78					
J3	D02	28					
J4	D03	77					
J5	D04	79					
J6	D05	80					
J7	D06	30					
J8	D07	81					
J9	D08	73					
J10	D09	72					
J11	D10	22					
J12	D11	71					
J13	D12	27					
J14	D13	74					
J15	D14	76					
J16	D15	26	-3.0	2.4	4.5	V	1-4
J17	PUO	2	-0.4	2.5	4.5	V	1-4
J18	PLO	6					
J19	EC1	53					
J20	EC2	3	-0.4	2.5	4.5	V	1-4

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TABLE X. Output voltage, high level, test - Continued.

Test cond	Signal name	Pin no.	Ioh (mA)	25 ± 5°C Initial limits Min	Max	Units	Notes
J21	EF1	57	-0.4	2.5	4.5	V	1-4
J22	CS0	95	-0.98	2.4	4.5	V	1-4
J23	CS1	85	-0.98	2.4	4.5	V	1-4
J24	UPDL	7	-0.38	2.4	4.5	V	1-4
J25	UPDU	5	-0.38	2.4	4.5	V	1-4
J26	PDL	9	-1.98	2.4	4.5	V	1-4
J27	PDU	60	-1.98	2.4	4.5	V	1-4
J28	PCKL	58	-1.96	2.4	4.5	V	1-4
J29	PCKU	61	-1.96	2.4	4.5	V	1-4
J30	LPCB	98	-4.0	2.4	4.5	V	1-4
J31	UPCB	87	-4.0	2.4	4.5	V	1-4
J32	PCL	50	-0.39	2.4	4.5	V	1-4
J33	PCU	20	-0.39	2.4	4.5	V	1-4
J34	PGL	100	-0.375	2.5	4.5	V	1-4
J35	PGU	70	-0.375	2.5	4.5	V	1-4
J36	UWE	33	-0.37	2.5	4.5	V	1-4
J37	LWE	97	-0.37	2.5	4.5	V	1-4
J38	BOE	24	-0.36	2.5	4.5	V	1-4
J39	DIR	32	-0.31	2.5	4.5	V	1-4
J40	ROE	89	-0.96	2.5	4.5	V	1-4

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NOTES:

1. Output voltage, high level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	4.5 V	± 45 mV	J1-J40
Power supply current limit	1.275 A	NA	J1-J40
Forcing current, (Ioh)	-3.0 mA	± 30.0 uA	J1-J16
	-0.4 mA	± 4.0 uA	J17-J21
	-0.98 mA	± 9.8 uA	J22,J23
	-0.38 mA	± 3.8 uA	J24,J25
	-1.98 mA	± 19.8 uA	J26,J27
	-1.96 mA	± 19.6 uA	J28,J29
	-4.0 mA	± 40.0 uA	J30,J31
	-0.39 mA	± 3.9 uA	J32,J33
	-0.375 mA	± 3.75 uA	J34,J35
	-0.37 mA	± 3.7 uA	J36,J37
	-0.36 mA	± 3.6 uA	J38
	-0.31 mA	± 3.1 uA	J39
	-0.96 mA	± 9.6 uA	J40
Pattern voltage high, (VH)	2.0 V	± 20 mV	J1-J40
low, (VL)	0.8 V	± 8 mV	J1-J40
Output voltage measurement	NA	± 105 mV	J1-J16
	NA	± 100 mV	J17-J21
	NA	± 105 mV	J22-J33
	NA	± 100 mV	J34-J40

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NOTES: - Continued.

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

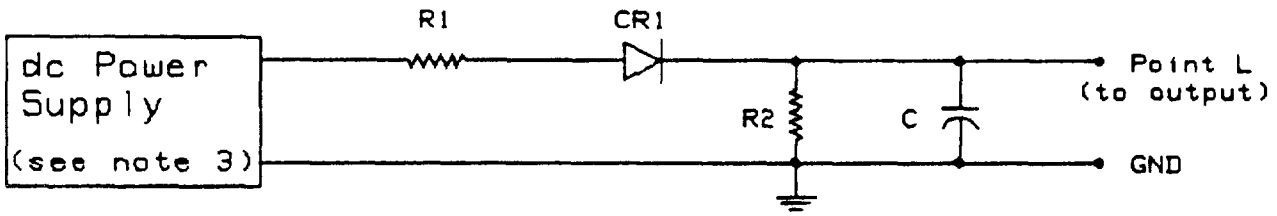
(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing current shall be applied to the pin specified in the appropriate test conditions in table X and the voltage then measured.

3. The load shall be as indicated on figure 8.

4. Abbreviations used are defined in 6.1.

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NOTES:

1. CR1 = 1N4148 or equivalent.

2. The load capacitance is defined as follows:

(a) C = 150 ± 15 pF (including probe and parasitic capacitance) for sample dynamic tests.

(b) C = 50 ± 5 pF (including probe and parasitic capacitance) for crosstalk and disable time tests.

(c) C = 300 ± 30 pF (including probe and parasitic capacitance) for qualification dynamic tests.

(d) The functional test shall be performed with only a 150 ± 15 pF load (including probe and parasitic capacitance) attached to each output.

(e) C not required for any other tests.

3. The load power supply voltage shall be 2.1 V ± 1 percent for the output disable time test and 2.5 V ± 1 percent for all other tests.

FIGURE 8. Logic load circuit.

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NOTES: - Continued.

4. Select R1 for a current flow of Iol when point L is held at Vol with the load power supply set at 2.5 V ± 1 percent. The values of Iol and Vol for each output are in the table below.

5. The value of R2 for each output load shall be as specified below.

Pin number	R2 1/ (ohms)	Iol (mA)	Vol (volts)
29,78,28,77,79, 80,30,81,73,72, 22,71,27,74,76, 26	800	12.0	0.4
2,6,53,3,57	6250	4.0	0.4

1/ All resistors are metal film, ± 1 percent, 1/4 W.

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TABLE XI. Output voltage, low level, test.

Test cond	Signal name	Pin no.	Iol (mA)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
K1	D00	29	12.0	0.0	0.4	V	1-4
K2	D01	78					
K3	D02	28					
K4	D03	77					
K5	D04	79					
K6	D05	80					
K7	D06	30					
K8	D07	81					
K9	D08	73					
K10	D09	72					
K11	D10	22					
K12	D11	71					
K13	D12	27					
K14	D13	74					
K15	D14	76					
K16	D15	26	12.0	0.0	0.4	V	1-4
K17	PU0	2	4.0	0.0	0.4	V	1-4
K18	PL0	6					
K19	EC1	53					
K20	EC2	3	4.0	0.0	0.4	V	1-4

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TABLE XI. Output voltage, low level, test - Continued.

Test cond	Signal name	Pin no.	Iol (mA)	25 ± 5°C Initial limits Min	Max	Units	Notes
K21	EF1	57	4.0	0.0	0.4	V	1-4
K22	CS0	95	19.98	0.0	0.5	V	1-4
K23	CS1	85	19.98	0.0	0.5	V	1-4
K24	UWE	33	3.98	0.0	0.4	V	1-4
K25	LWE	97	3.98	0.0	0.4	V	1-4
K26	PCL	50	3.75	0.0	0.4	V	1-4
K27	PCU	20	3.75	0.0	0.4	V	1-4
K28	PDL	9	7.6	0.0	0.5	V	1-4
K29	PDU	60	7.6	0.0	0.5	V	1-4
K30	PGL	100	3.75	0.0	0.4	V	1-4
K31	PGU	70	3.75	0.0	0.4	V	1-4
K32	PCKL	58	7.6	0.0	0.5	V	1-4
K33	PCKU	61	7.6	0.0	0.5	V	1-4
K34	LPCB	98	7.75	0.0	0.4	V	1-4
K35	UPCB	87	7.75	0.0	0.4	V	1-4
K36	UPDL	7	3.6	0.0	0.4	V	1-4
K37	UPDU	5	↓	↓	↓	↓	↓
K38	BOE	24	3.6	0.0	0.4	V	1-4
K39	DIR	32	1.6	0.0	0.4	V	1-4
K40	ROE	89	19.96	0.0	0.5	V	1-4

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:07Z
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NOTES:

1. Output voltage, low level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	4.5 V	± 45 mV	K1-K40
Power supply current limit	1.275 A	NA	K1-K40
Forcing current, (Iol)	12.0 mA	± 120.0 uA	K1-K16
	4.0 mA	± 40.0 uA	K17-K21
	19.98 mA	± 199.0 uA	K22,K23
	3.98 mA	± 39.8 uA	K24,K25
	3.75 mA	± 37.5 uA	K26,K27
	7.6 mA	± 76.0 uA	K28,K29
	3.75 mA	± 37.5 uA	K30,K31
	7.6 mA	± 76.0 uA	K32,K33
	7.75 mA	± 77.5 uA	K34,K35
	3.6 mA	± 36.0 uA	K36-K38
	1.6 mA	± 16.0 uA	K39
	19.96 mA	± 199.0 uA	K40
Pattern voltage high, (VH)	3.0 V	± 100 mV	K1-K40
low, (VL)	0.0 V	± 100 mV	K1-K40
Output voltage measurement	NA	± 20 mV	K1-K21
	NA	± 25 mV	K22,K23
	NA	± 20 mV	K24-K27
	NA	± 25 mv	K28,K29
	NA	± 20 mV	K30,K31
	NA	± 25 mV	K32,K33
	NA	± 20 mV	K34-K39
	NA	± 25 mV	K40

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NOTES: - Continued.

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing current shall be applied to the pin specified in the appropriate test conditions in table XI and the voltage then measured.

3. The load shall be as indicated on figure 8.

4. Abbreviations used are defined in 6.1.

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TABLE XII. Output leakage current, high level, test.

Test cond	Signal name	Pin no.	Forcing voltage (V)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
L1	D00	29	2.7	-20	20	uA	1-3
L2	D01	78					
L3	D02	28					
L4	D03	77					
L5	D04	79					
L6	D05	80					
L7	D06	30					
L8	D07	81					
L9	D08	73					
L10	D09	72					
L11	D10	22					
L12	D11	71					
L13	D12	27					
L14	D13	74					
L15	D14	76					
L16	D15	26	2.7	-20	20	uA	1-3

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NOTES:

1. Output leakage current, high level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	L1-L16
Power supply current limit	1.275 A	NA	L1-L16
Forcing voltage, (Voh)	2.7 V	± 135 mV	L1-L16
Pattern voltage high, (VH) low, (VL)	3.0 V 0.0 V	± 100 mV ± 100 mV	L1-L16 L1-L16
Output current measurement	NA	± 2 uA	L1-L16

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

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NOTES: - Continued.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table XII and the current then measured. The forcing voltage overshoot shall not exceed 100 mV.

3. Abbreviations used are defined in 6.1.

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TABLE XIII. Output leakage current, low level, test.

Test cond	Signal name	Pin no.	Vol (V)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
M1	D00	29	0.4	-200	200	uA	1-3
M2	D01	78					
M3	D02	28					
M4	D03	77					
M5	D04	79					
M6	D05	80					
M7	D06	30					
M8	D07	81					
M9	D08	73					
M10	D09	72					
M11	D10	22					
M12	D11	71					
M13	D12	27					
M14	D13	74					
M15	D14	76					
M16	D15	26	0.4	-200	200	uA	1-3

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NOTES:

1. Output leakage current, low level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	M1-M16
Power supply current limit	1.275 A	NA	M1-M16
Forcing voltage, (Vol)	0.4 V	± 20 mV	M1-M16
Pattern voltage high, (VH) low, (VL)	3.0 V 0.0 V	± 100 mV ± 100 mV	M1-M16 M1-M16
Output current measurement	NA	± 20 uA	M1-M16

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

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NOTES: - Continued.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table XIII and the current then measured. The forcing voltage overshoot shall not exceed 100 mV.

3. Abbreviations used are defined in 6.1.

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TABLE XIV. Power supply current test.

Test cond	Parameter	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes
N	Icc	1,31	81.8	818.5	mA	1-3

NOTES:

1. Power supply current test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	N
Power supply current limit	1.275 A	NA	N
Pattern voltage high, (VH) low, (VL)	3.0 V 0.0 V	± 100 mV ± 100 mV	N N
Power supply current measurement	NA	± 36.8 mA	N

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test and take a measurement at the pattern line with the maximum power supply current as indicated in table XIV.

3. Abbreviations used are defined in 6.1.

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TABLE XV. Functional test.

Test cond	Test description	Notes
P1	Functional test minimum power supply voltage maximum pattern rate	1-3
P2	Functional test maximum power supply voltage maximum pattern rate	1-3

NOTES:

1. Functional test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	4.5 V 5.5 V	± 45 mV ± 55 mV	P1 P2
Power supply current limit	1.275 A	NA	P1,P2
Pattern voltage high, (VH) low, (VL)	 2.4 V 0.4 V	 ± 120 mV ± 20 mV	 P1,P2 P1,P2
Output voltage verification high, (Voh) high, (Voh) low, (Vol)	 NA NA NA	 ± 105 mV ± 155 mV ± 25 mV	 P1 P2 P1,P2

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NOTES: - Continued.

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Use the appropriate pattern to verify the functional performance of the module as indicated in table XV using only a capacitive load of 100 pF (see figure 8). The output compare levels shall be as follows:

High level: $2.4\ V \leq Voh \leq Vcc$
Low level: $0.0\ V \leq Vol \leq 0.4\ V$

(c) The module inputs shall not violate the minimum input timing parameters referenced on figure 9.

3. Abbreviations used are defined in 6.1.

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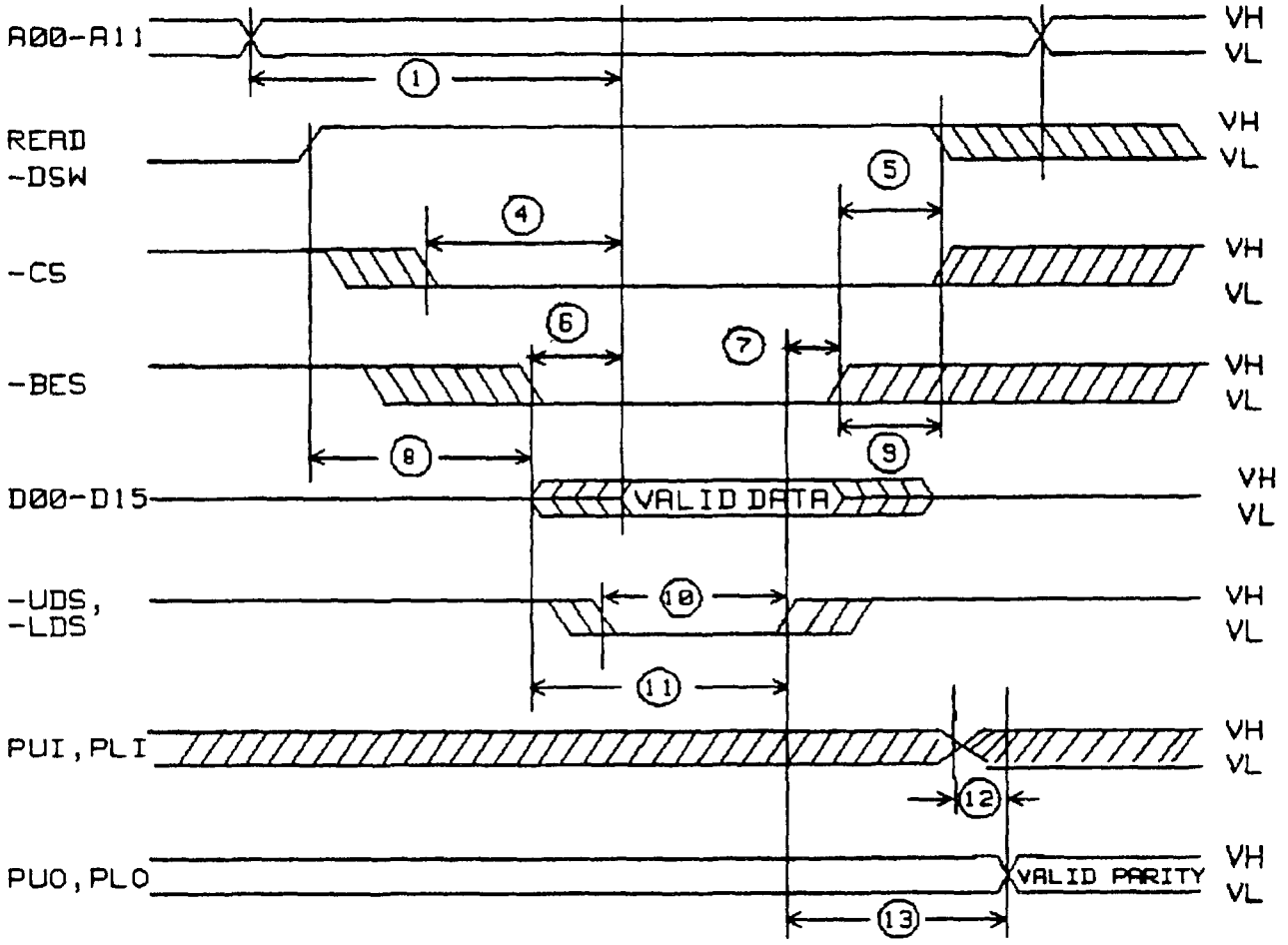


FIGURE 9a. Input timing parameters for table XV (read cycle).

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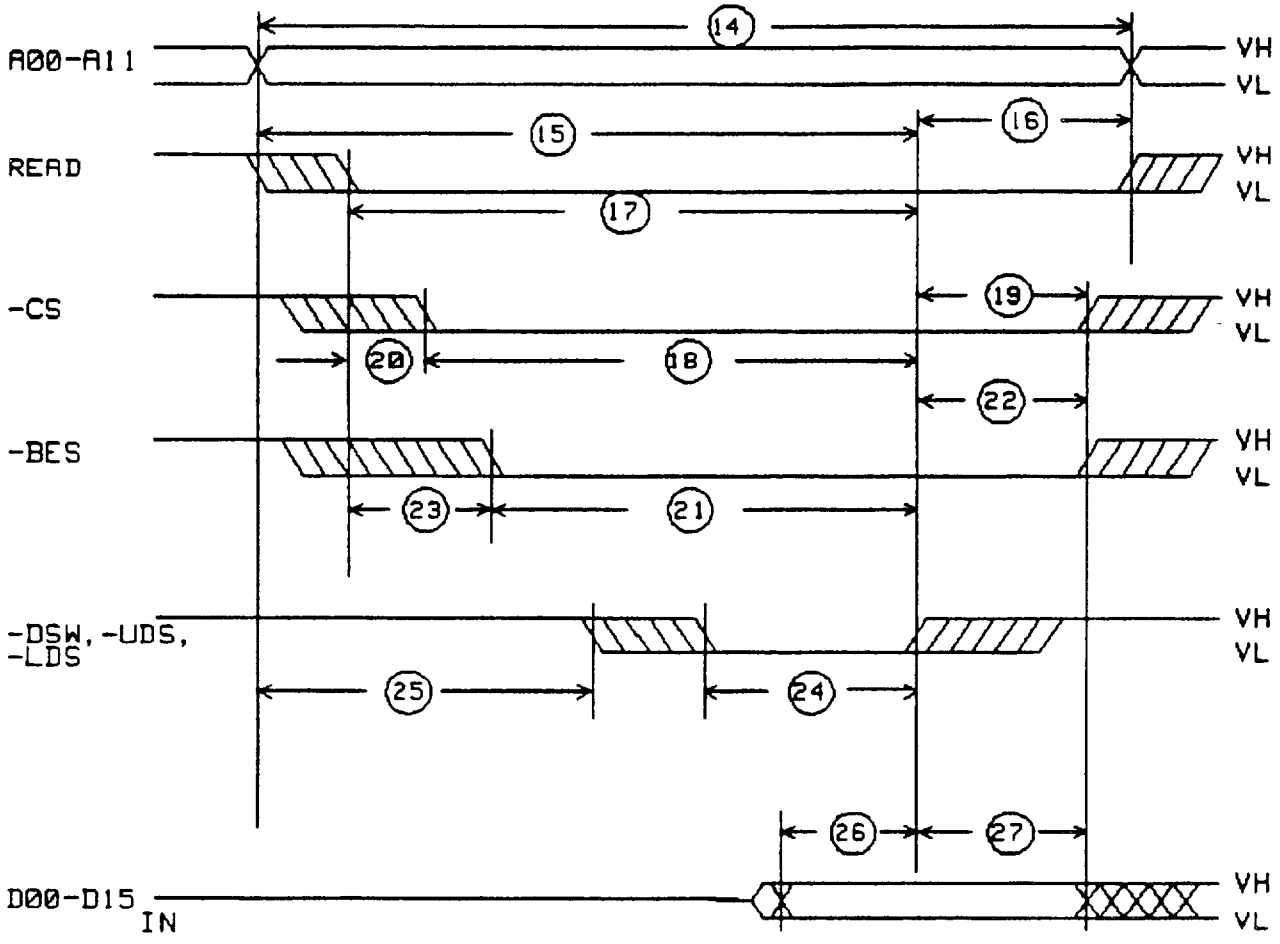


FIGURE 9b. Input timing parameters for table XV (write cycle).

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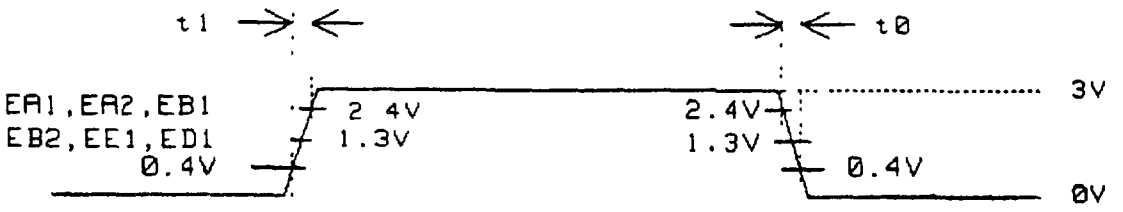


FIGURE 9c. Input timing parameters for table XV (spare gates).

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NOTES:

1. 193 nS minimum.
4. 175 nS minimum.
5. 0 nS minimum.
6. 119 nS minimum.
7. 5 nS minimum.
8. 24 nS minimum.
9. 54 nS minimum.
10. 25 nS minimum.
11. 221 nS minimum.
12. 47 nS minimum.
13. 150 nS minimum.
14. 323 nS minimum.
15. 114 nS minimum.
16. 45 nS minimum.
17. 278 nS minimum.
18. 214 nS minimum.
19. 45 nS minimum.
20. 0 nS minimum.
21. 200 nS minimum.
22. 45 nS minimum.
23. 78 nS minimum.
24. 55 nS minimum.
25. 39 nS minimum.
26. 129 nS minimum.
27. 45 nS minimum.

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TABLE XVI. Dynamic test.

Test cond	Test description	Notes
Q1	Dynamic test, sample maximum pattern rate minimum power supply	1-7
Q2	Dynamic test, sample maximum pattern rate maximum power supply	1-7
Q3	Dynamic test, qualification maximum pattern rate minimum power supply	1-7
Q4	Dynamic test, qualification maximum pattern rate maximum power supply	1-7

NOTES:

1. Dynamic test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	4.5 V 5.5 V	± 45 mV ± 55 mV	Q1,Q3 Q2,Q4
Power supply current limit	1.275 A	NA	Q1-Q4
Input voltage high, (VH) low, (VL)	 2.4 V 0.4 V	 ± 120 mV ± 20 mV	 Q1-Q4 Q1-Q4

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NOTES: - Continued.

Condition	Value	Accuracy	Test condition
Output voltage verification			
high, (Voh)	NA	± 105 mV	Q1,Q3
high, (Voh)	NA	± 155 mV	Q2,Q4
low, (Vol)	NA	± 25 mV	Q1-Q4

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) The output compare levels shall be as follows:

High level: $2.4 \leq Voh \leq Vcc$
Low level: $0.0\ V \leq Vol \leq 0.4\ V$

3. The load circuit for this test is specified on figure 8.

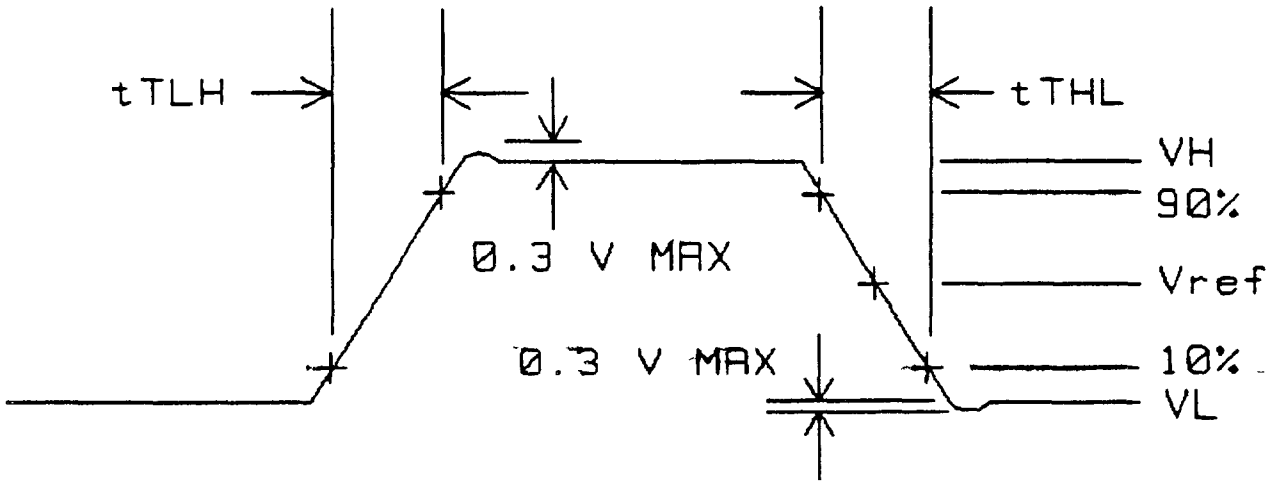
4. All of the input waveforms shall meet the requirements shown on figure 10.

5. This test shall be performed as indicated in table XVI. The dynamic parameters are found in table XVII.

6. Input/Output (I/O) waveforms are as referenced on figure 11.

7. Abbreviations used are defined in 6.1.

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NOTES:

1. VH and VL are as specified in the test tolerance table.
2. Vref = 1.3 V.
3. Rise and fall times for the above waveform are as follows:

Input pin	tTLH	tTHL
64,65,68,12,14,13,17,19, 67,16,66,21	15.0 ± 1.5 nS	6.0 ± 1.0 nS
29,78,28,77,79,80,30,81, 73,72,22,71,27,74,76,26	15.0 ± 1.5 nS	6.0 ± 1.0 nS
90	2.5 ± 0.25 nS	2.5 ± 0.25 nS
18	2.5 ± 0.25 nS	2.5 ± 0.25 nS
59,69,63,62,55,56,23,99, 08,11,51,52,54,04	15.0 ± 1.5 nS	6.0 ± 1.0 nS

FIGURE 10. Sample input waveform for table XVI.

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TABLE XVII. Dynamic parameters.

Ref	Parameter number	Parameter	0 to 60°C		25 ± 5°C Initial limits		Units	Fig
			Min	Max	Min	Max		
29	1-2	EA1,EB1 to EC1 (tPLH)	0	47	0	47	nS	11a
28	3-4	EA1,EB1 to EC1 (tPHL)	0	47	0	47	nS	11a
29	5-6	EA2,EB2 to EC2 (tPLH)	0	47	0	47	nS	11a
28	7-8	EA2,EB2 to EC2 (tPHL)	0	47	0	47	nS	11a
29	9-10	EE1,ED1 to EF1 (tPLH)	0	52	0	52	nS	11a
28	11-12	EE1,ED1 to EF1 (tPHL)	0	52	0	52	nS	11a
12	13-14	PUI,PLI to parity (tPLH)	0	47	0	47	nS	11b
12	15-16	PUI,PLI to parity (tPHL)	0	47	0	47	nS	11b
13	17-18	-UDS,-LDS to parity (LH)	0	150	0	150	nS	11b
13	19-20	-UDS,-LDS to parity (HL)	0	150	0	150	nS	11b
10	21	tPW (parity clock pulse width)	25	-	25	-	nS	11b

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TABLE XVII. Dynamic parameters - Continued.

Ref	Parameter number	Parameter	0 to 60°C		25 ± 5°C Initial limits		Units	Fig
			Min	Max	Min	Max		
11	22	tPS (parity setup) -BES to parity clock EC1 (tPHL)	221	-	221	-	nS	11b
7	23	tPH (parity hold) parity clock to -BES	5	-	5	-	nS	11b
8	24	READ to -BES setup time (read cycle)	24	-	24	-	nS	11b
9	25	-BES to READ hold time (read cycle)	54	-	54	-	nS	11b
5	26	-BES to -CS hold time (read cycle)	0	-	0	-	nS	11b
15	27-38	A00-A11 to valid write	114	-	114	-	nS	11c
17	39	READ to valid write	278	-	278	-	nS	11c
18	40	-CS to valid write	214	-	214	-	nS	11c
21	41	-BES to valid write	200	-	200	-	nS	11c

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TABLE XVII. Dynamic parameters - Continued.

Ref	Parameter number	Parameter	0 to 60°C		25 ± 5°C Initial limits		Units	Fig
			Min	Max	Min	Max		
25	42-77	A00-A11 to -DSW,-UDS, -LDS (tas - address setup time)	39	-	39	-	nS	11c
20	78	READ to -CS (write cycle)	0	-	0	-	nS	11c
23	79	READ to -BES setup time (write cycle)	78	-	78	-	nS	11c
26	80	tDW - data setup time (data in to write pulse)	129	-	129	-	nS	11c
24	81	twp - -DSW, -UDS,-LDS write pulse width	55	-	55	-	nS	11c
27	82	tdh - data hold after write	45	-	45	-	nS	11c
16	83	tar - address hold after write	45	-	45	-	nS	11c
16	84	trh - READ hold after write	45	-	45	-	nS	11c

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TABLE XVII. Dynamic parameters - Continued.

Ref	Parameter number	Parameter	0 to 60°C		25 ± 5°C Initial limits		Units	Fig
			Min	Max	Min	Max		
19	85	tch - -CS hold after write (write cycle)	45	-	45	-	nS	11c
22	86	tBH - -BES hold after write	45	-	45	-	nS	11c
14	87	tWC - write cycle time	323	-	323	-	nS	11c
1	88-99	A00-A11 to D00-D15 (tPLH-taa)	0	179	0	179	nS	11b
1	100-111	A00-A11 to D00-D15 (tPHL-taa)	0	179	0	179	nS	11b
4	112-127	-CS to D00-D15 (tPZL)	0	154	0	154	nS	11b
4	128-143	-CS to D00-D15 (tPZH)	0	154	0	154	nS	11b
6	144-159	-BES to D00 -D15 (tPZL)	0	107	0	107	nS	11b
6	160-175	-BES to D00 -D15 (tPZH)	0	107	0	107	nS	11b
1	176-187	A00-A11 to D00-D15 (tPLH-taa)	0	193	0	193	nS	11b
1	188-199	A00-A11 to D00-D15 (tPHL-taa)	0	193	0	193	nS	11b

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TABLE XVII. Dynamic parameters - Continued.

Ref	Parameter number	Parameter	0 to 60°C		25 ± 5°C Initial limits		Units	Fig
			Min	Max	Min	Max		
4	200-215	-CS to D00-D15 (tPZL)	0	166	0	166	nS	11b
4	216-231	-CS to D00-D15 (tPZH)	0	166	0	266	nS	11b
6	232-247	-BES to D00 -D15 (tPZL)	0	119	0	119	nS	11b
6	248-263	-BES to D00 -D15 (tPZH)	0	119	0	119	nS	11b

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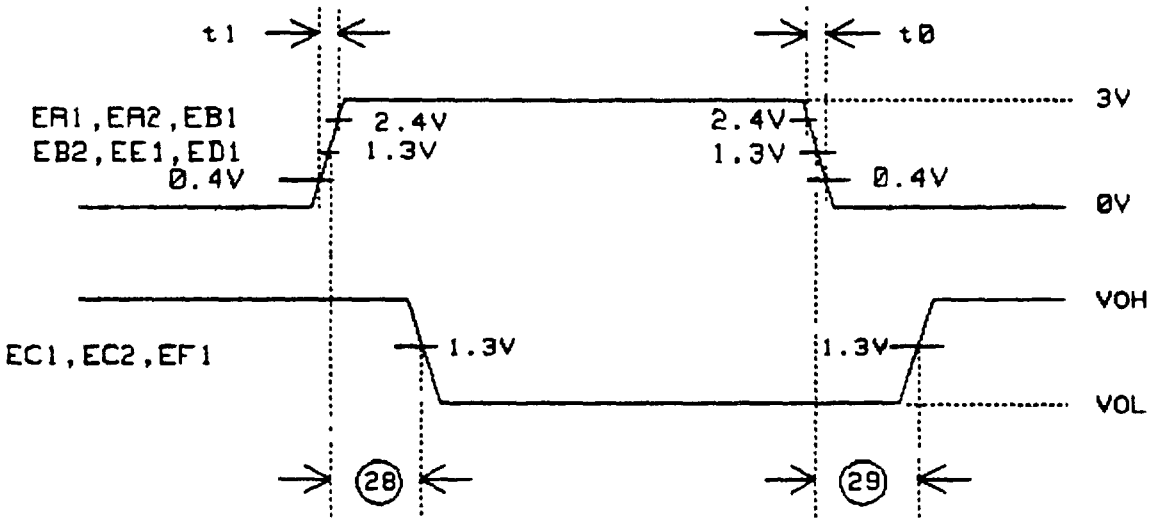


FIGURE 11a. Dynamic waveforms for table XVII (spare gates).

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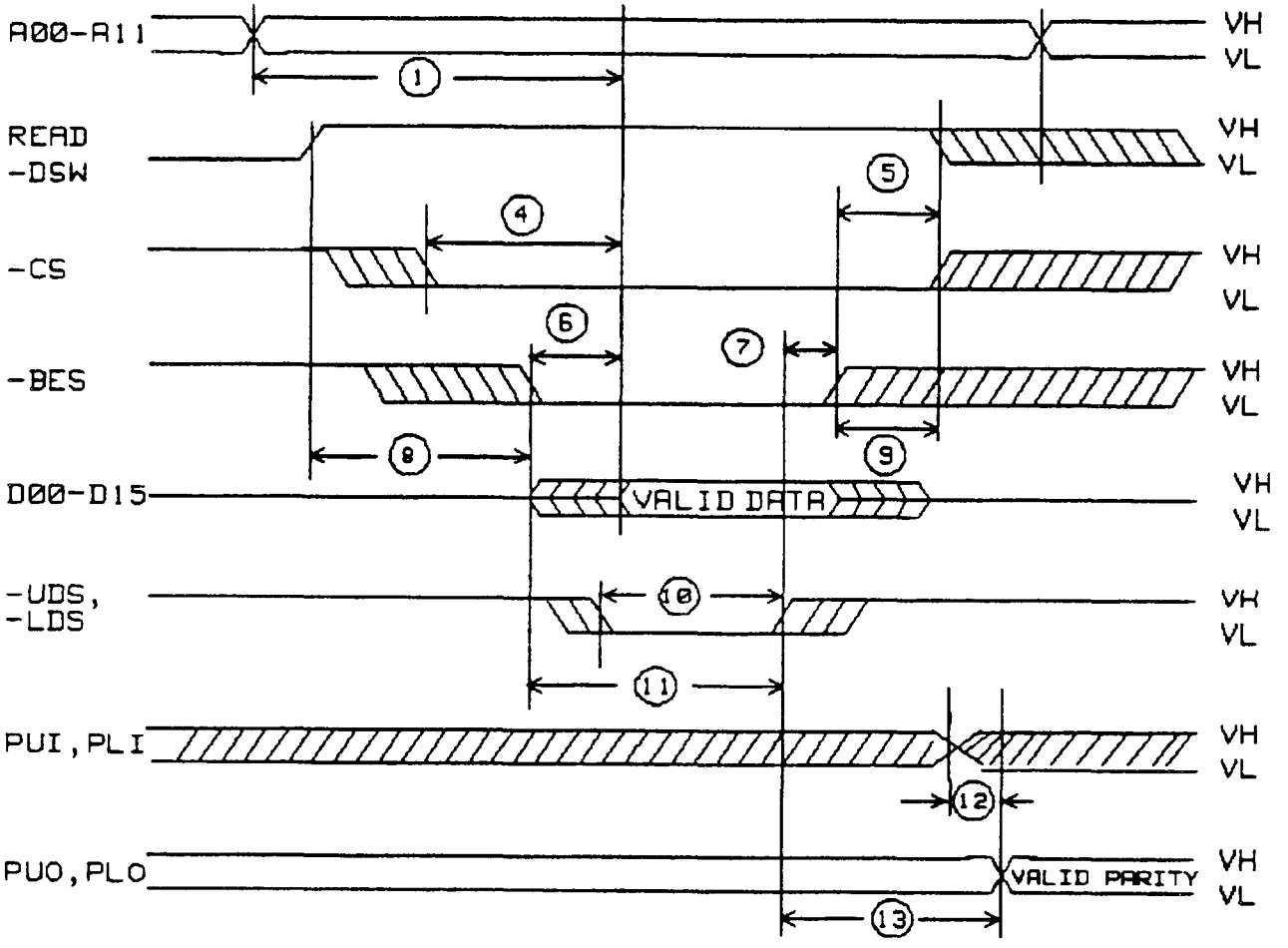


FIGURE 11b. Dynamic waveforms for table XVII (read cycle).

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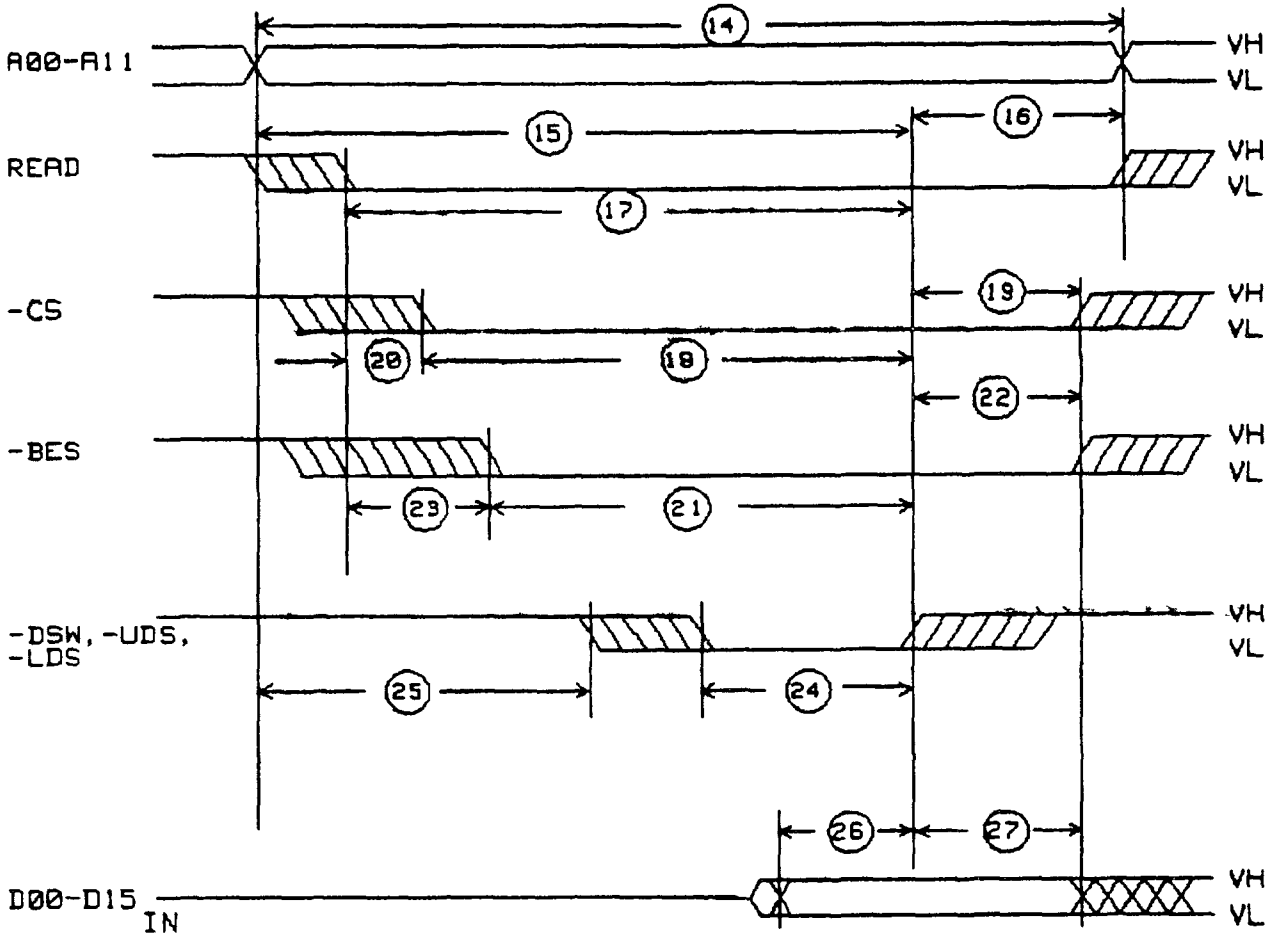


FIGURE 11c. Dynamic waveforms for table XVII (write cycle).

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TABLE XVIII. Power supply filter capacitance test.

Test cond	Parameter	Pin numbers	25 ± 5°C Initial limits Min	Max	Units	Notes
R	PSFC	1,31	3.78	7.02	uF	1-3

NOTES:

1. Power supply filter capacitance test tolerance table.

Condition	Value	Accuracy	Test condition
Capacitance bridge frequency	1 KHz	± 100 Hz	R
Capacitance bridge voltage	0.3 Vp	± 30 mVp	R
Capacitance measurement	NA	± 162 nF	R

2. The capacitance bridge used in this test shall be item C01. The negative terminal shall be connected to the module ground and the positive terminal to the pin indicated in table XVIII.

3. Abbreviations used are defined in 6.1.

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TABLE XIX. Power supply transient amplitude test.

Test cond	Parameter	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes
S	PSTA	1	0	400	mV	1-3

NOTES:

1. Power supply transient amplitude test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.0 V	± 250 mV	S
Power supply current limit	1.275 A	NA	S
Pattern voltage high, (VH) low, (VL)	 3.0 V 0.0 V	 ± 100 mV ± 100 mV	 S S
Power supply transient amplitude measurement	NA	± 20 mV	S

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table to an external source. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

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NOTES: - Continued.

(b) Precondition the module under test and make the indicated measurement for the test condition shown in table XIX.

(c) All Vcc pins shall be shorted together and the scope point, probe 'A', shall be connected to the pin indicated in table XIX.

(d) Using the oscilloscope, D06, inspect for the most negative spike as shown on figure 12 at the pattern line with the maximum power supply current.

(e) The maximum probe capacitance is 15 pF. There shall be no other capacitance present on Vcc to ground.

3. Abbreviations used are defined in 6.1.

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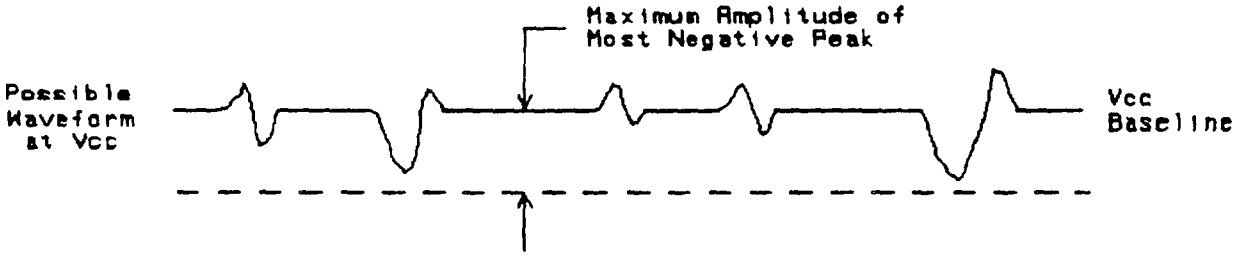


FIGURE 12. Power supply transient amplitude waveform for table XIX.

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TABLE XX. Output crosstalk, high level, test.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes	Figs
T1	EC1	53	0	400	mV	1-5	13-15
T2	EC2	3					
T3	EF1	57					
T4	PUO	2					
T5	PLO	6	0	400	mV	1-5	13-15

NOTES:

1. Output crosstalk, high level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.0 V	± 0.25 V	T1-T5
Power supply current limit	1.275 A	NA	T1-T5
Pattern voltage high, (VH) low, (VL)	 3.0 V 0.0 V	 ± 100 mV ± 100 mV	 T1-T5 T1-T5
Output crosstalk measurement	NA	± 20 mV	T1-T5

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NOTES: - Continued.

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) Use probe 'A' to verify the pattern voltage and pattern repetition rate of 1000 nS ± 200 nS, 50 percent duty cycle.

(d) For each output under test in table XX, connect probe 'B' and visually inspect the output for the most negative spike as indicated on figure 13.

(e) The maximum probe capacitance is 10 pF.

3. The test circuit shall be as indicated on figure 14. The input waveforms are shown on figure 15.

4. All the outputs shall be loaded as specified on figure 8.

5. Abbreviations used are defined in 6.1.

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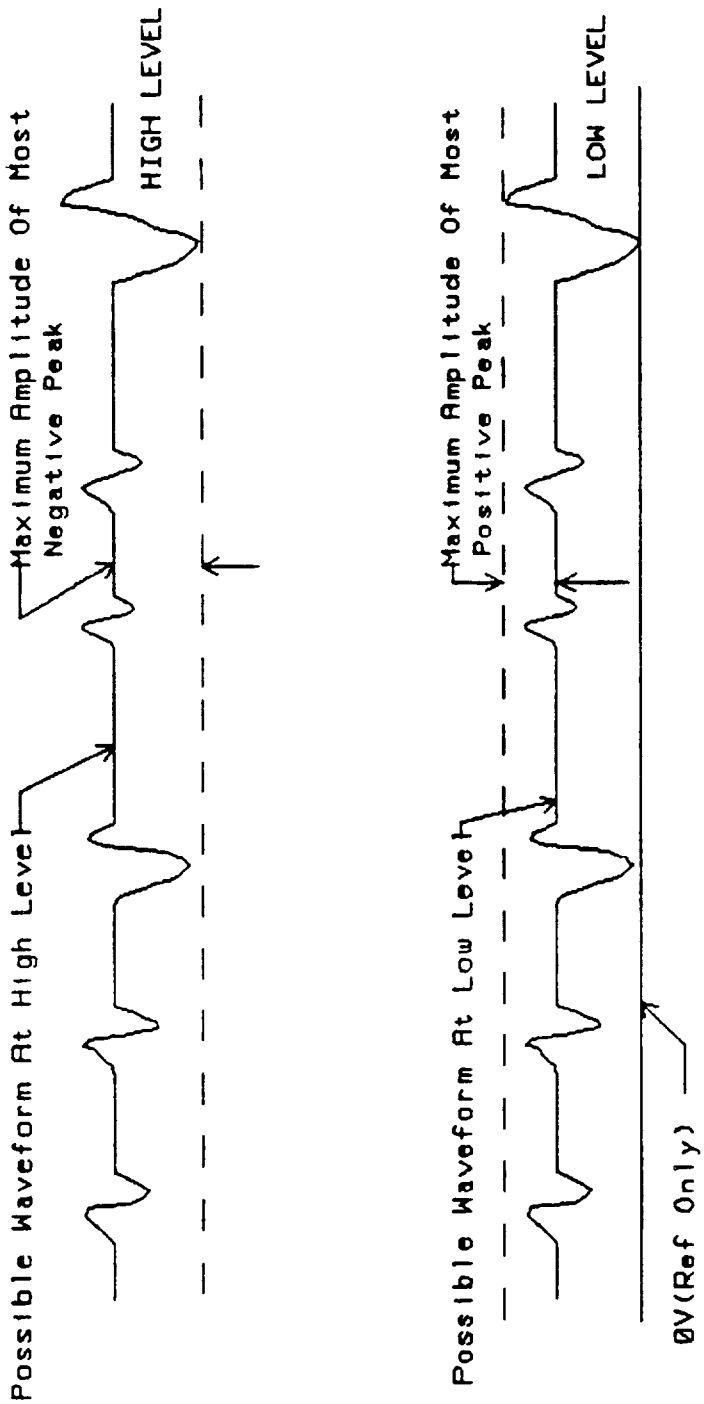


FIGURE 13. Output crosstalk, high and low level, test waveforms for tables XX and XXI.

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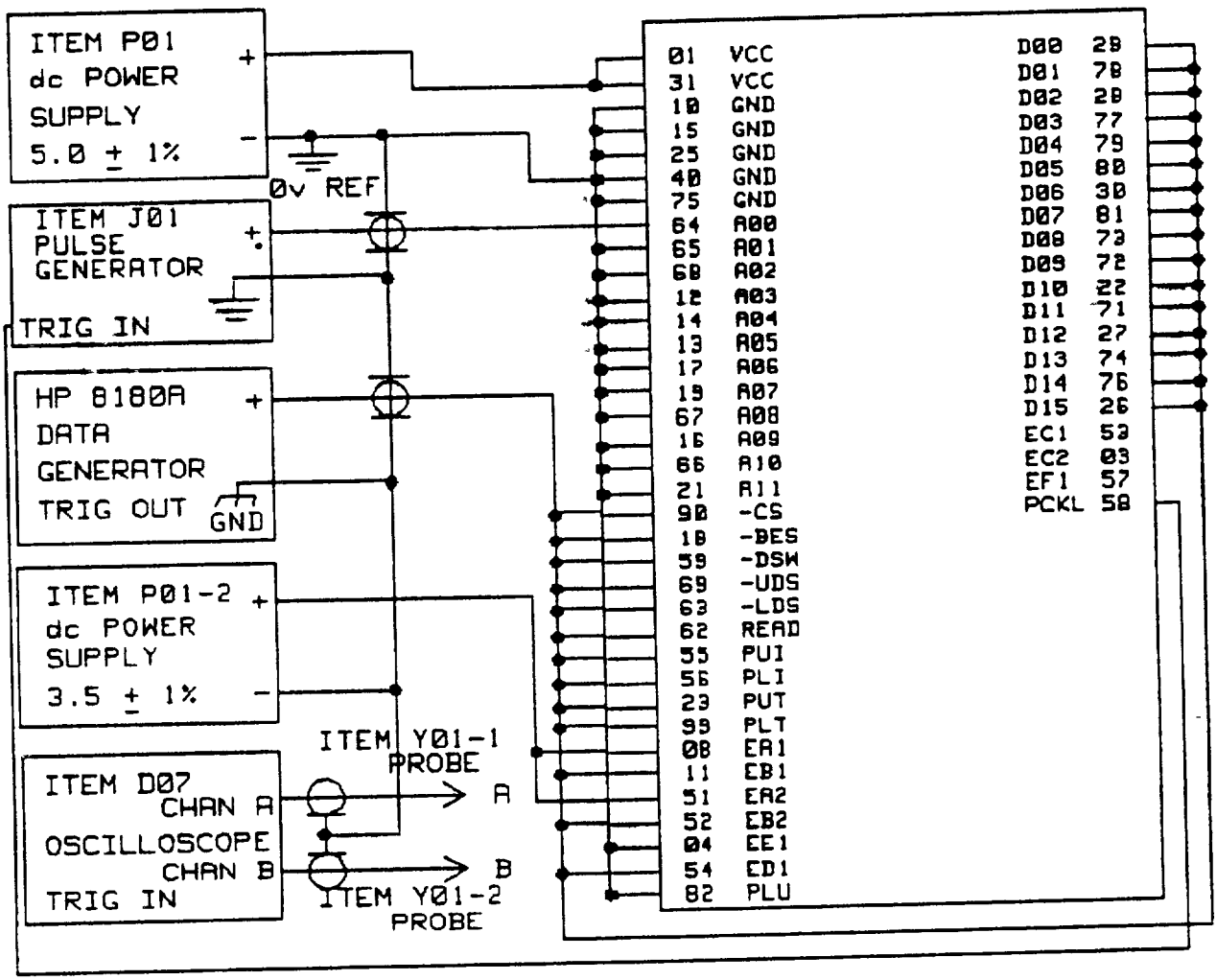
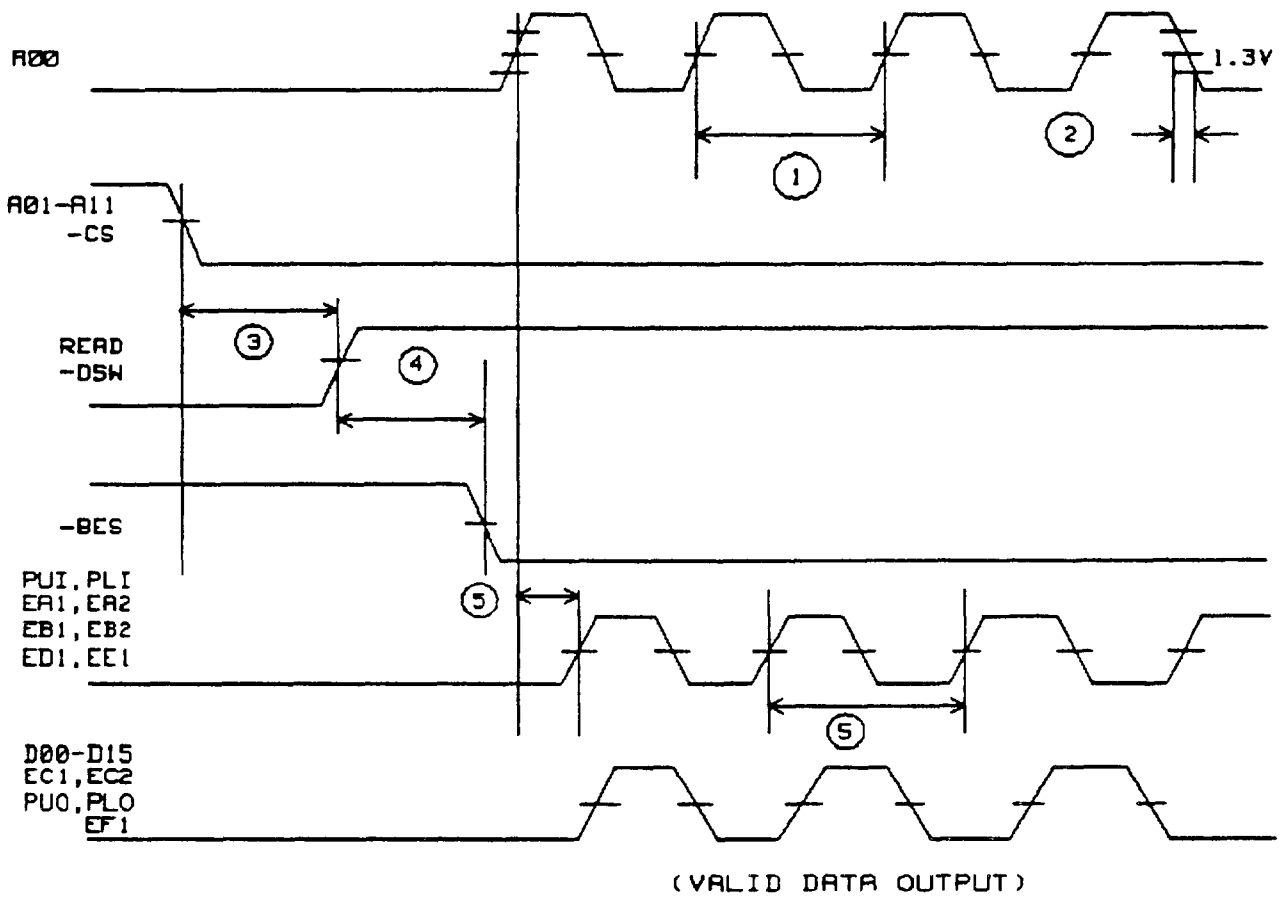


FIGURE 14. Output crosstalk, high and low level, test circuit for tables XX and XXI.

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NOTES:

1. Address A00 cycle time equals 400 nS ± 10 nS.

2. Tr = Tf = 6 nS ± 1.5 nS.

3. Setup time equals 20 nS, min.

4. Setup time equals 20 ns, min.

5. Adjust these values to obtain a signal skew of less than 3 nS between D00-D15 and the following outputs: EC1, EC2, EF1, PU0, and PL0.

FIGURE 15. Input waveforms for tables XX and XXI.

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TABLE XXI. Output crosstalk, low level, test.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes	Figs
U1	EC1	53	0	400	mV	1-5	13-15
U2	EC2	3					
U3	EF1	57					
U4	PUO	2					
U5	PLO	6	0	400	mV	1-5	13-15

NOTES:

1. Output crosstalk, low level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.0 V	± 0.25 V	U1-U5
Power supply current limit	1.275 A	NA	U1-U5
Pattern voltage high, (VH) low, (VL)	 3.0 V 0.0 V	 ± 100 mV ± 100 mV	 U1-U5 U1-U5
Output crosstalk measurement	NA	± 20 mV	U1-U5

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NOTES: - Continued.

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) Use probe 'A' to verify the pattern voltage and pattern repetition rate of 1000 nS ± 200 nS, 50 percent duty cycle.

(d) For each output under test in table XXI, connect probe 'B' and visually inspect the output for the most positive spike as indicated on figure 13.

(e) The maximum probe capacitance is 10 pF.

3. The test circuit shall be as indicated on figure 14. The input waveforms are shown on figure 15.

4. All outputs shall be loaded as specified on figure 8.

5. Abbreviations used are defined in 6.1.

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TABLE XXII. Input capacitance test.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
V1	A00	64	2	30	pF	1-3
V2	A01	65				
V3	A02	68				
V4	A03	12				
V5	A04	14				
V6	A05	13				
V7	A06	17				
V8	A07	19				
V9	A08	67				
V10	A09	16				
V11	A10	66				
V12	A11	21	2	30	pF	1-3
V13	D00	29	2	60	pF	1-3
V14	D01	78				
V15	D02	28				
V16	D03	77				
V17	D04	79				
V18	D05	80				
V19	D06	30				
V20	D07	81	2	60	pF	1-3

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TABLE XXII. Input capacitance test - Continued.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes
V21	D08	73	2	60	pF	1-3
V22	D09	72				
V23	D10	22				
V24	D11	71				
V25	D12	27				
V26	D13	74				
V27	D14	76				
V28	D15	26				
V29	-CS	90				
V30	-BES	18				
V31	-DSW	59				
V32	-UDS	69				
V33	-LDS	63				
V34	READ	62				
V35	PUI	55				
V36	PLI	56				
V37	PUT	23				
V38	PLT	99	2	60	pF	1-3
V39	EA1	8	2	30	pF	1-3
V40	EB1	11	2	30	pF	1-3

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TABLE XXII. Input capacitance test - Continued.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes
V41	EA2	51	2	30	pF	1-3
V42	EB2	52				
V43	ED1	54				
V44	EE1	4	2	30	pF	1-3
V45	PLU	82	2	75	pF	1-3

NOTES:

1. Input capacitance test tolerance table.

Condition	Value	Accuracy	Test condition
Capacitance meter frequency	1 MHz	± 100 KHz	V1-V45
Capacitance meter voltage	15 mV	± 1.5 mV	V1-V45
Capacitance measurement	NA NA NA NA	± 1.4 pF ± 2.9 pF ± 1.4 pF ± 3.65 pF	V1-V12 V13-V38 V39-V44 V45

2. The capacitance meter used in this test shall be item C05. The negative terminal shall be connected to the module ground and the positive terminal to the pin indicated in table XXII.

3. Abbreviations used are defined in 6.1.

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TABLE XXIII. Output capacitance test.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
W1	LWE	97	0	60	pF	1-3
W2	UWE	33				
W3	CS0	95				
W4	CS1	85				
W5	PDL	9				
W6	PDU	60				
W7	PCKL	58				
W8	PCKU	61				
W9	PCL	50				
W10	PCU	20				
W11	PGL	100				
W12	PGU	70				
W13	UPDL	7				
W14	UPDU	5				
W15	BOE	24				
W16	DIR	32				
W17	UPCB	87				
W18	LPCB	98				
W19	ROE	89				
W20	PUO	2	0	60	pF	1-3

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TABLE XXIII. Output capacitance test - Continued.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes
W21	PLO	6	0	30	pF	1-3
W22	EC1	53				
W23	EC2	3				
W24	EF1	57	0	30	pF	1-3

NOTES:

1. Output capacitance test tolerance table.

Condition	Value	Accuracy	Test condition
Capacitance meter frequency	1 MHz	± 100 KHz	W1-W24
Capacitance meter voltage	15 mV	± 1.5 mV	W1-W24
Capacitance measurement	NA NA	± 3.0 pF ± 1.5 pF	W1-W20 W21-W24

2. The capacitance meter used in this test shall be item C05. The negative terminal shall be connected to the module ground and the positive terminal to the pin indicated in table XXIII.

3. Abbreviations used are defined in 6.1.

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TABLE XXIV. Disable time test.

Test cond	Signal name	Connect A to pin	Connect B to pin	Connect C to pin	25 + 5°C Initial limits Min	25 + 5°C Initial limits Max	Units	Figs
Low to high impedance (tPLZ)								
X1	-CS to D00	Open	32	29	6	88	nS	16, 17
X2	-CS to D01			78				
X3	-CS to D02			28				
X4	-CS to D03			77				
X5	-CS to D04			79				
X6	-CS to D05			80				
X7	-CS to D06			30				
X8	-CS to D07			81				
X9	-CS to D08			73				
X10	-CS to D09			72				
X11	-CS to D10			22				
X12	-CS to D11			71				
X13	-CS to D12			27				
X14	-CS to D13			74				
X15	-CS to D14	Open	32	76	6	88	nS	16, 17

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TABLE XXIV. Disable time test - Continued.

Test cond	Signal name	Connect A to pin	Connect B to pin	Connect C to pin	25 + 5°C Initial limits Min	25 + 5°C Initial limits Max	Units	Figs
X16	-CS to D15	Open	32	26	6	88	nS	16,17
High to high impedance (tPHZ)								
X17	-CS to D00	Open	32	29	6	88	nS	16,17
X18	-CS to D01			78				
X19	-CS to D02			28				
X20	-CS to D03			77				
X21	-CS to D04			79				
X22	-CS to D05			80				
X23	-CS to D06			30				
X24	-CS to D07			81				
X25	-CS to D08			73				
X26	-CS to D09			72				
X27	-CS to D10			22				
X28	-CS to D11			71				
X29	-CS to D12			27				
X30	-CS to D13	Open	32	74	6	88	nS	16,17

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TABLE XXIV. Disable time test - Continued.

Test cond	Signal name	Connect A to pin	Connect B to pin	Connect C to pin	25 + 5°C Initial limits Min	25 + 5°C Initial limits Max	Units	Figs
X31	-CS to D14	Open	32	76	6	88	nS	16,17
X32	-CS to D15	Open	32	26	6	88	nS	16,17
Low to high impedance (tPLZ)								
X33	-BES to D00	Open	32	29	4	74	nS	16,17
X34	-BES to D01			78				
X35	-BES to D02			28				
X36	-BES to D03			77				
X37	-BES to D04			79				
X38	-BES to D05			80				
X39	-BES to D06			30				
X40	-BES to D07			81				
X41	-BES to D08			73				
X42	-BES to D09			72				
X43	-BES to D10			22				
X44	-BES to D11			71				
X45	-BES to D12	Open	32	27	4	74	nS	16,17

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TABLE XXIV. Disable time test - Continued.

Test cond	Signal name	Connect A to pin	Connect B to pin	Connect C to pin	25 + 5°C Initial limits Min	25 + 5°C Initial limits Max	Units	Figs
X46	-BES to D13	Open	32	74	4	74	nS	16, 17
X47	-BES to D14			76				
X48	-BES to D15	Open	32	26	4	74	nS	16, 17
High to high impedance (tPHZ)								
X49	-BES to D00	Open	32	29	4	74	nS	16, 17
X50	-BES to D01			78				
X51	-BES to D02			28				
X52	-BES to D03			77				
X53	-BES to D04			79				
X54	-BES to D05			80				
X55	-BES to D06			30				
X56	-BES to D07			81				
X57	-BES to D08			73				
X58	-BES to D09			72				
X59	-BES to D10			22				
X60	-BES to D11	Open	32	71	4	74	nS	16, 17

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TABLE XXIV. Disable time test - Continued.

Test cond	Signal name	Connect A to pin	Connect B to pin	Connect C to pin	25 + 5°C Initial limits Min	25 + 5°C Initial limits Max	Units	Figs
X61	-BES to D12	Open	32	27	4	74	nS	16, 17
-X62	-BES to D13			74				
X63	-BES to D14			76				
X64	-BES to D15	Open	32	26	4	74	nS	16, 17

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NOTES:

1. Disable time test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply	5.0 V	± 50 mV	X1-X64
Power supply current limits	1.275 A	NA	X1-X64
Pattern voltage high, (VH) low, (VL)	 3.0 V 0.0 V	 ± 100 mV ± 100 mV	 X1-X64 X1-x64
Disable time measurements	 NA NA	 ± 4.10 nS ± 3.50 nS	 X1-X32 X33-X64

2. The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

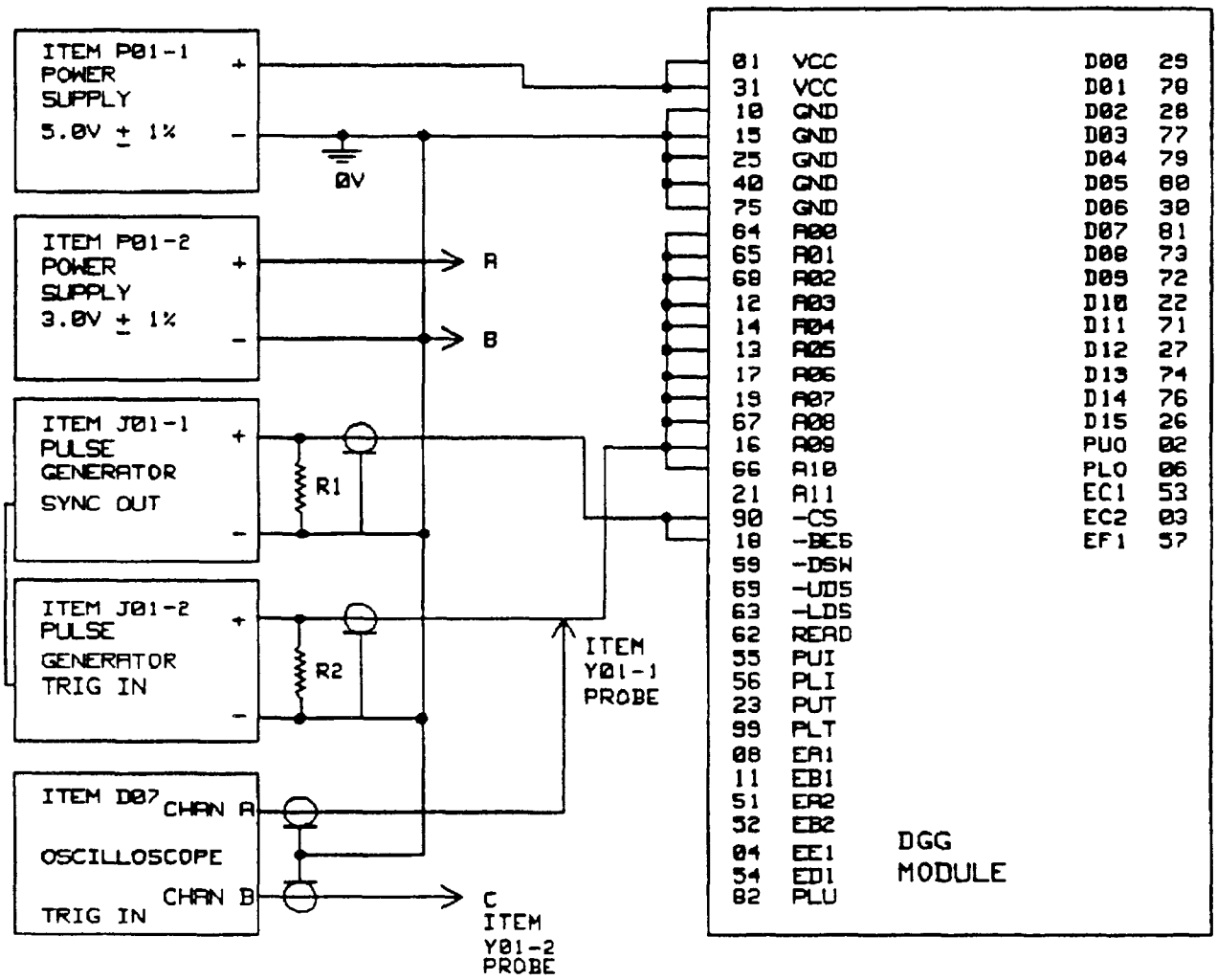
3. The test circuit shall be as indicated on figure 16.

4. Module inputs shall meet the timing requirements shown on figure 17.

5. Module outputs shall be loaded as specified on figure 8.

6. Abbreviations used are defined in 6.1.

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NOTE:

1. R1 = R2 = 51 ohms ± 5 percent, 1/2 W.

FIGURE 16. Disable time test circuit for table XXIV.

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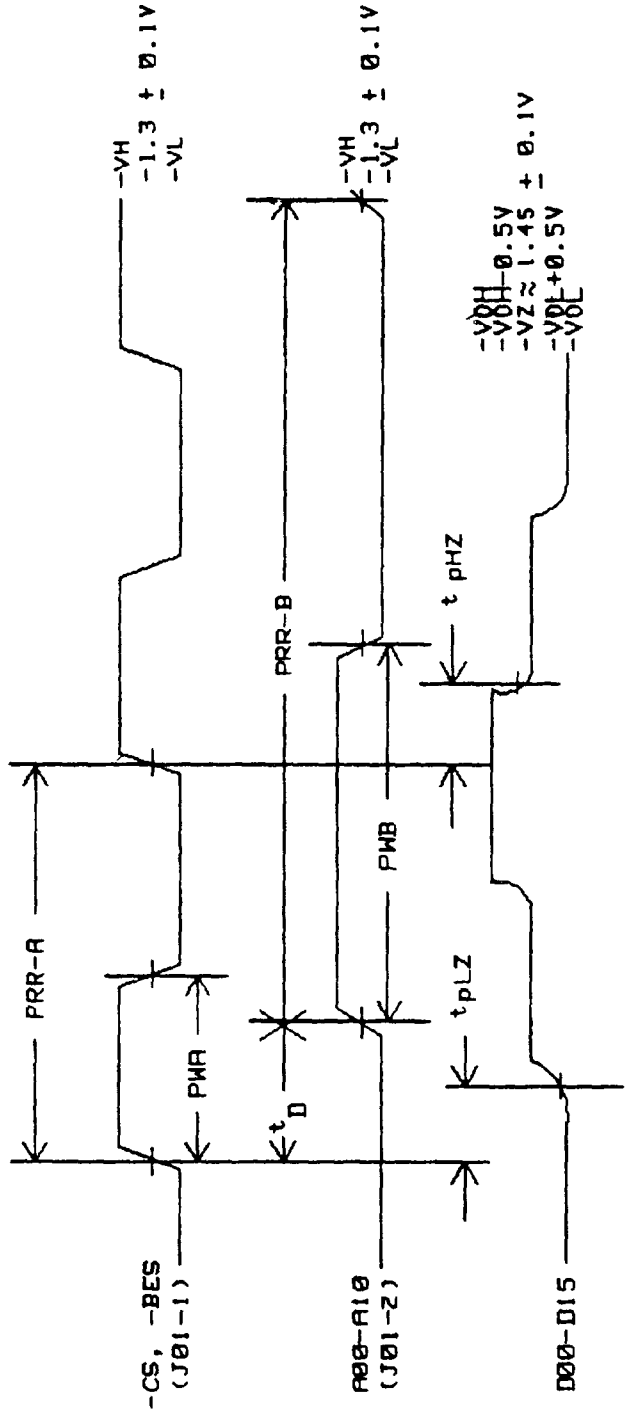


FIGURE 17. Disable time test waveforms for table XXIV.

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NOTES:

1. PWA = 250 ± 25 nS.
   PRR-A = 500 ± 50 nS.

2. PWB = 500 ± 50 nS.
   PRR-B = 1000 ± 100 nS.

3. tp = 125 ± 10 nS.

4. Rise and fall times shall be 2.5 ± 1.0 nS.

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TABLE XXV. Threshold test.

Test cond	Parameter	Notes	Fig
Y	Threshold	1-4	9

NOTES:

1. Threshold test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	4.5 V	± 45 mV	Y
Power supply current limit	1.275 A	NA	Y
Pattern voltage high, (VH) low, (VL)	 2.0 V 0.7 V	 ± 20 mV ± 7 mV	 Y Y
Output voltage measurement high, (Voh) high, (Voh) low, (Vol)	 NA NA NA	 ± 105 mV ± 100 mV ± 25 mV	 Y Y Y

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

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NOTES: - Continued.

(b) Precondition the module under test and verify the output patterns by monitoring the outputs and comparing the results to the output low and high levels specified in the output voltage tests. The module inputs shall not violate the minimum input timing parameters referenced on figure 9.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

4. Abbreviations used are defined in 6.1.

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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-M-28787/366	24 August 1990

3. DOCUMENT TITLE
Modules. Digital, Standard Electrojic Static Random Access Memory Module, Key Code DGG

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

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James Reeves, (Code 06QM2	(703) 746-1926	
c. ADDRESS (Include Zip Code) Commander, Naval Sea Systems Command Department of the Navy Washington, DC 20362-5101	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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